

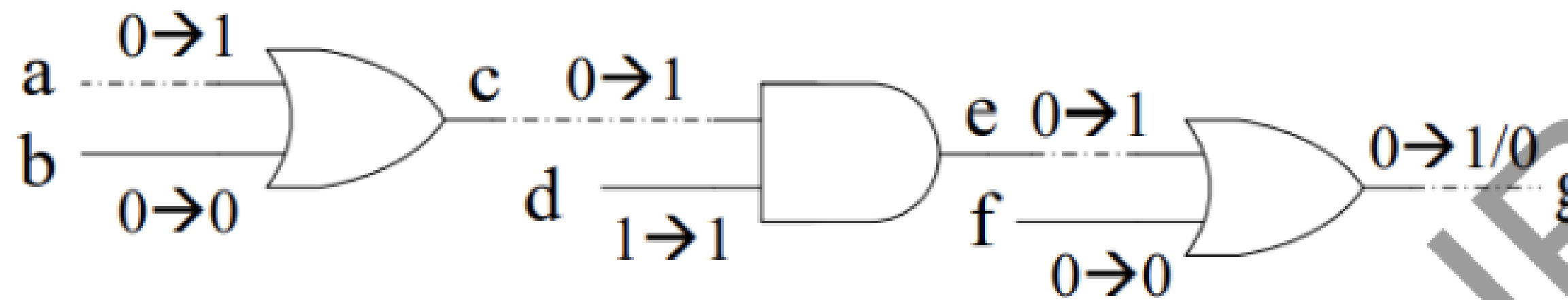
Path Delay Fault model

PATH DELAY FAULT MODEL

- Associated with testing at-speed performance of critical path (given by designer).
- Longest path or the path whose clock frequency is high is usually defined as critical path because timing violations are more likely to happen.
- Path delay fault model :
Path delay of faulty path $>$ clock period
- Path delay fault model is used to detect some manufacturing defect that is distributed throughout the path (Eg : more resistive path)

DIFFERENCE BETWEEN PATH DELAY FAULT MODEL AND TDF

- **Path Delay Fault model** : Assume that fault is distributed along the path.

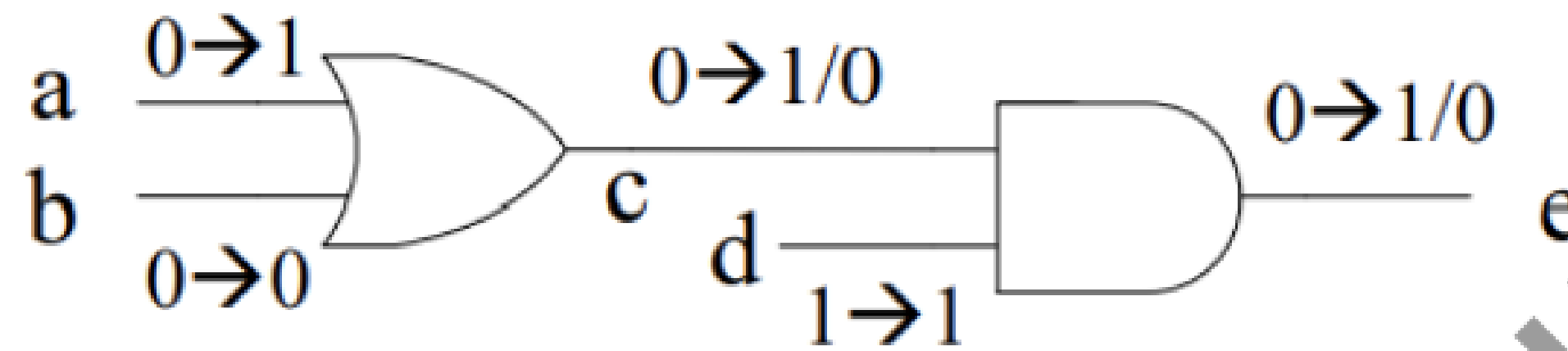


A test for the path delay fault associated with path a-c-e-g

- Path delay of faulty path > clock period
- Therefore path delay fault model can be used to detect some manufacturing defect that is distributed throughout the path (Eg : more resistive path)
(The illustrations is given in the next slides)

DIFFERENCE BETWEEN PATH DELAY FAULT MODEL AND TDF

- **Transition Delay Fault model** : Assume that fault is large and lumped (specific to a particular node).



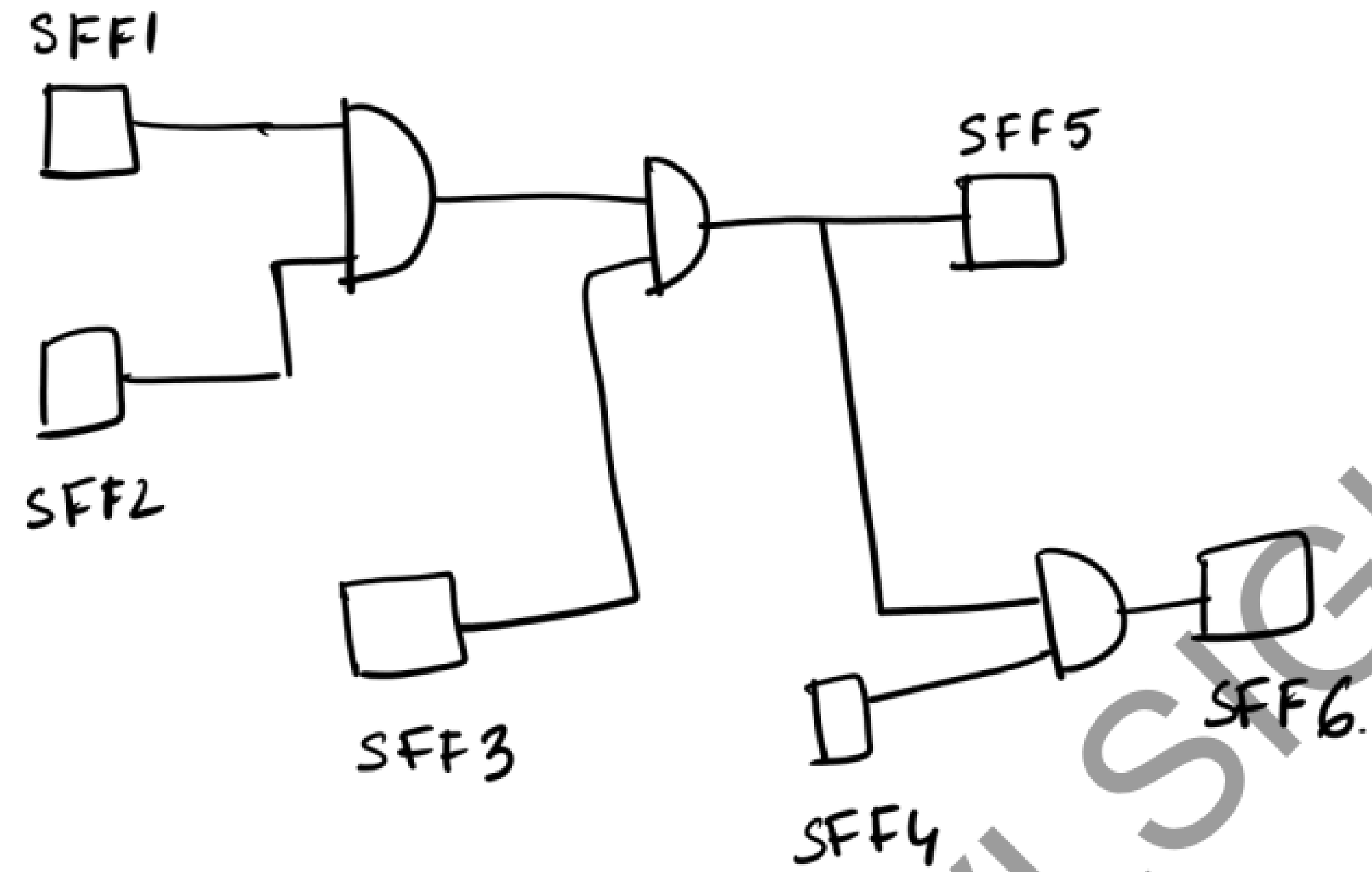
A test for the slow-to-rise transition fault at c

- Therefore transition delay fault model can be used to detect some manufacturing defect at a particular node. (The illustrations is given in the next slides)

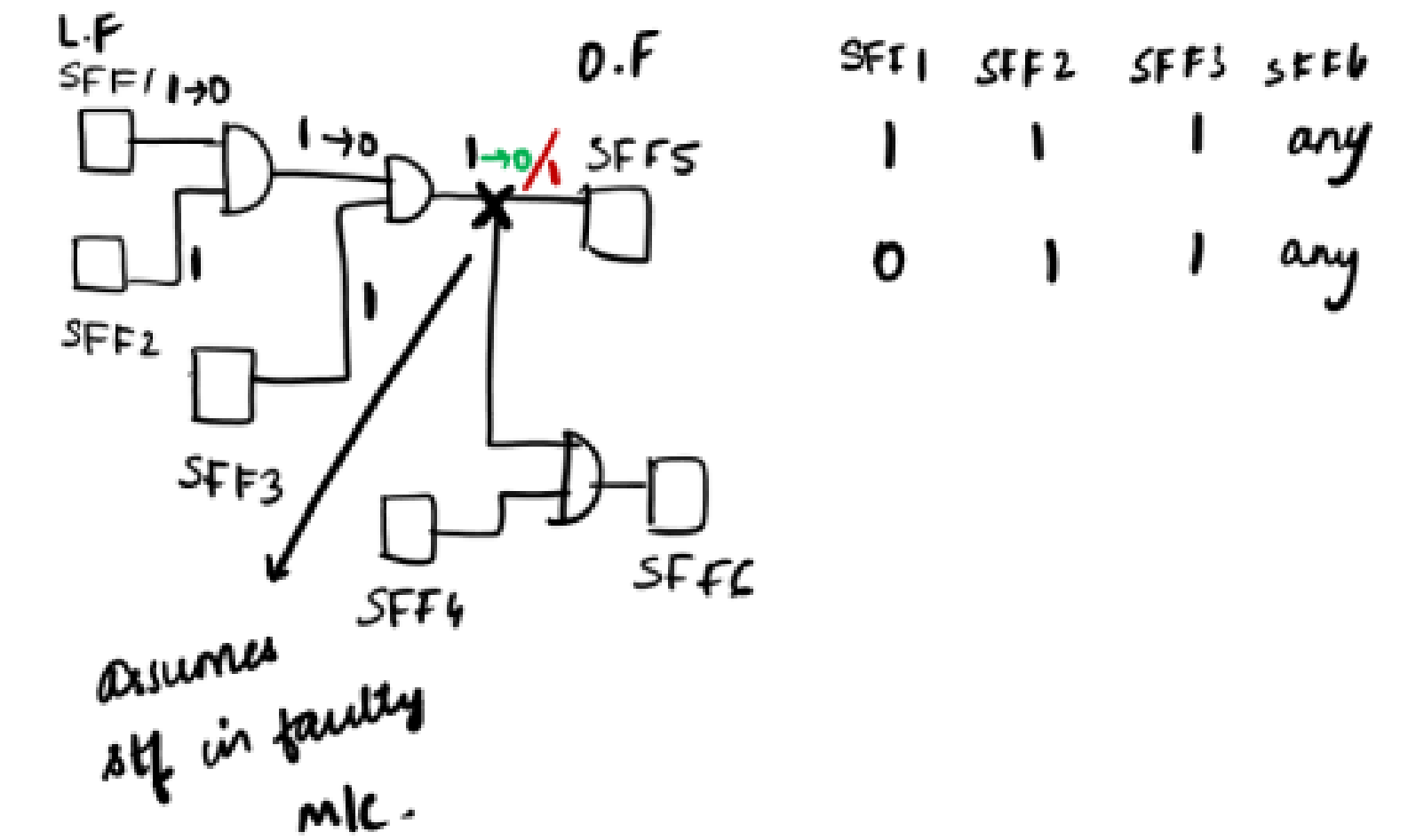
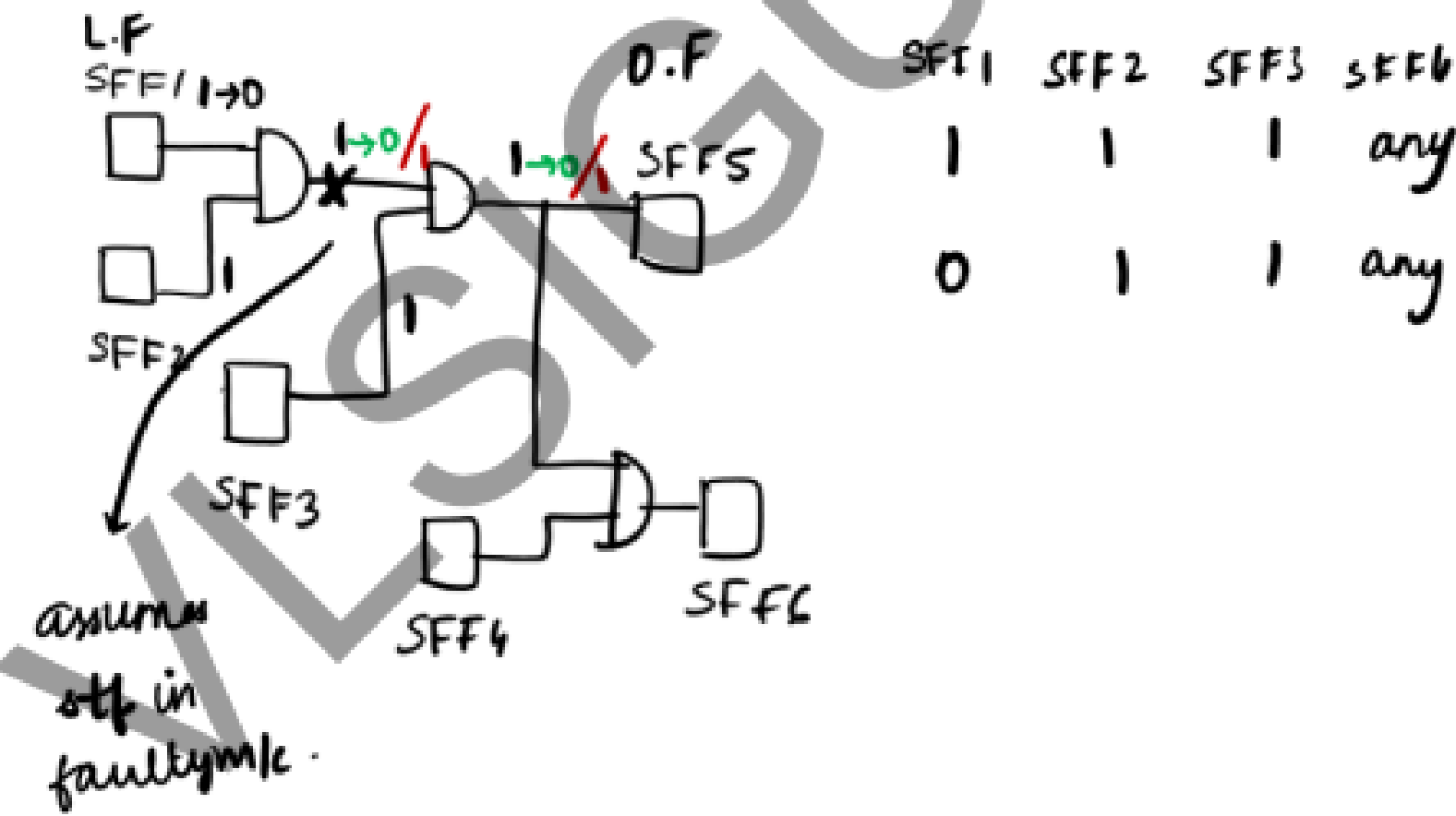
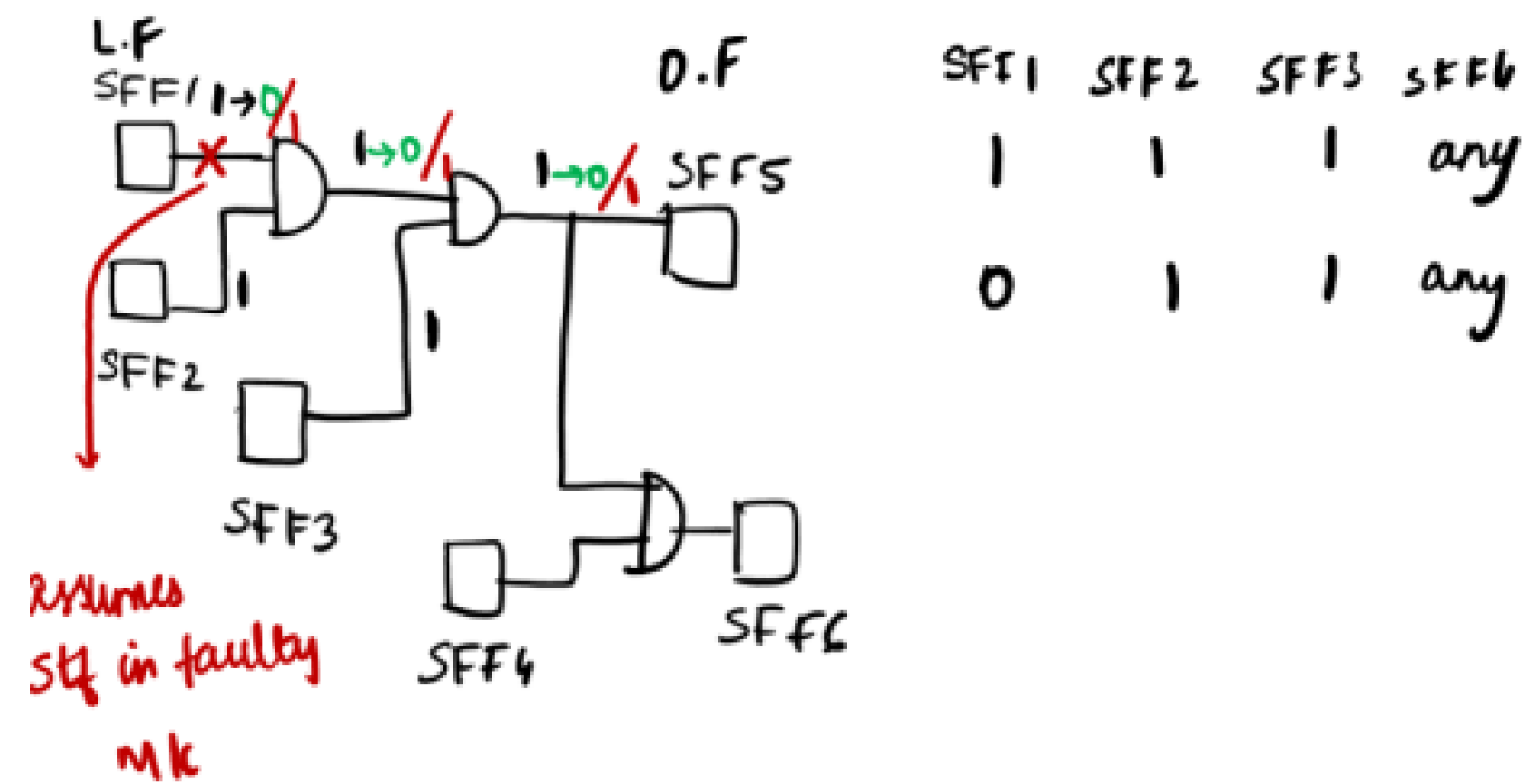
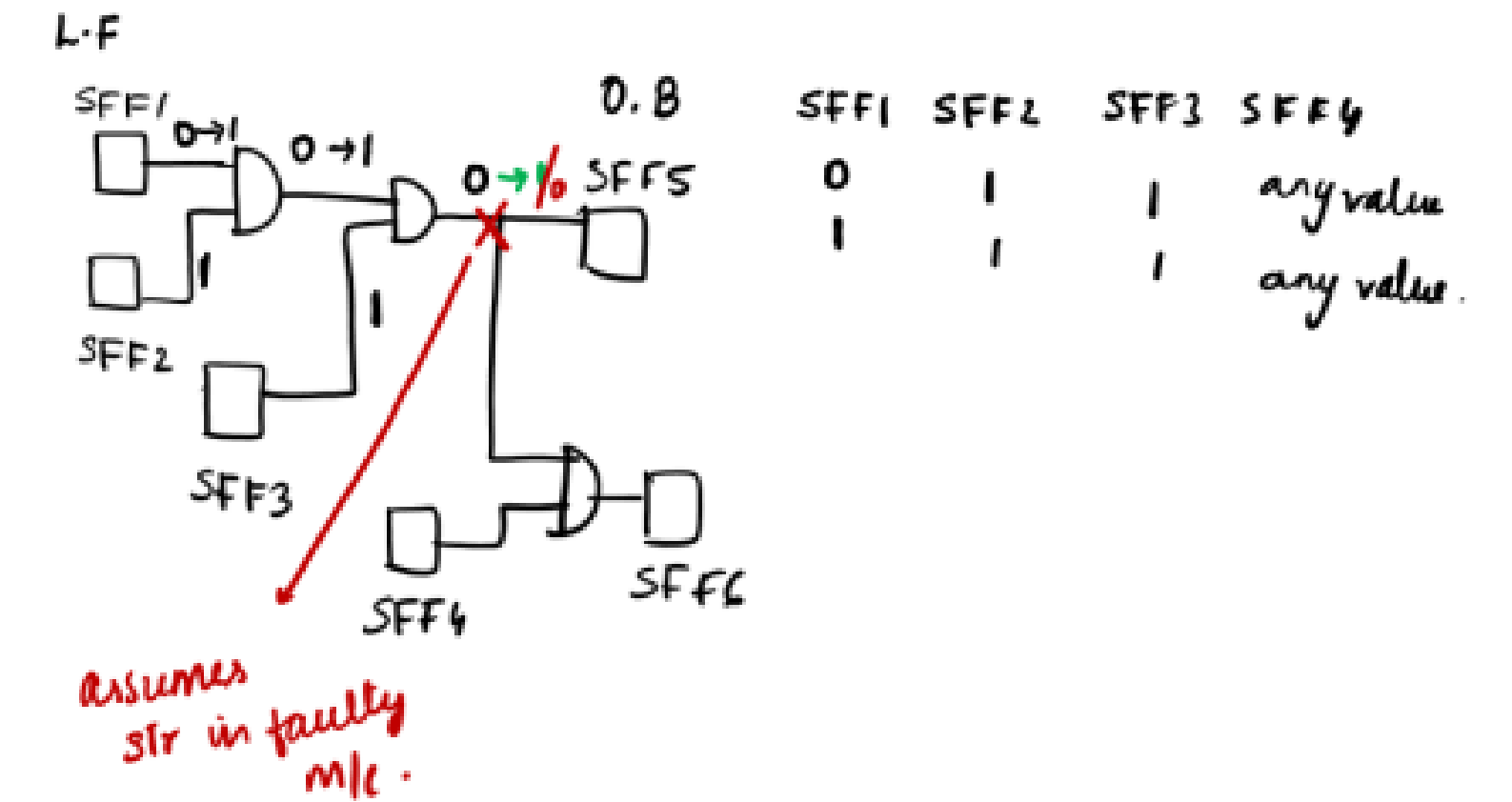
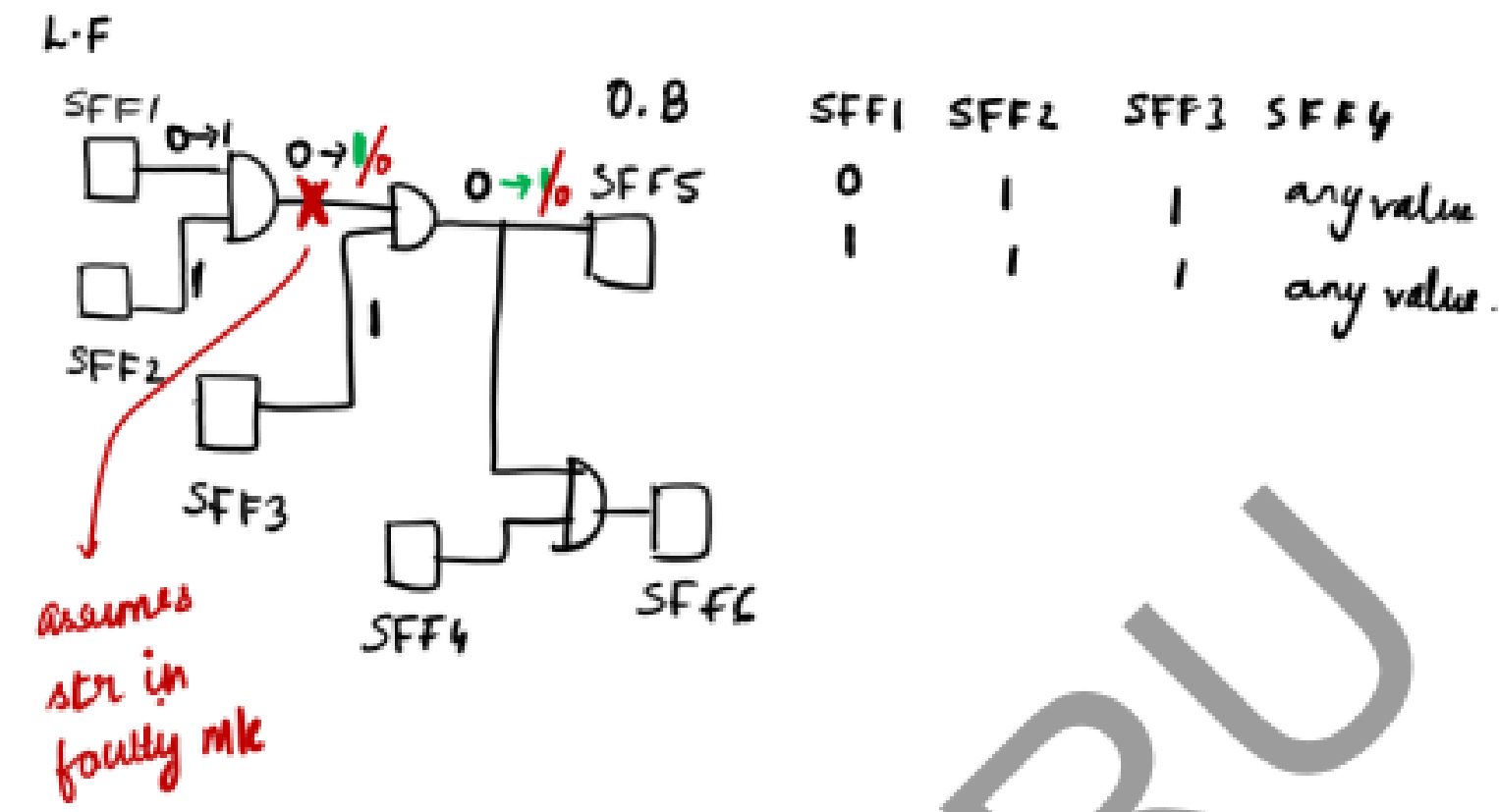
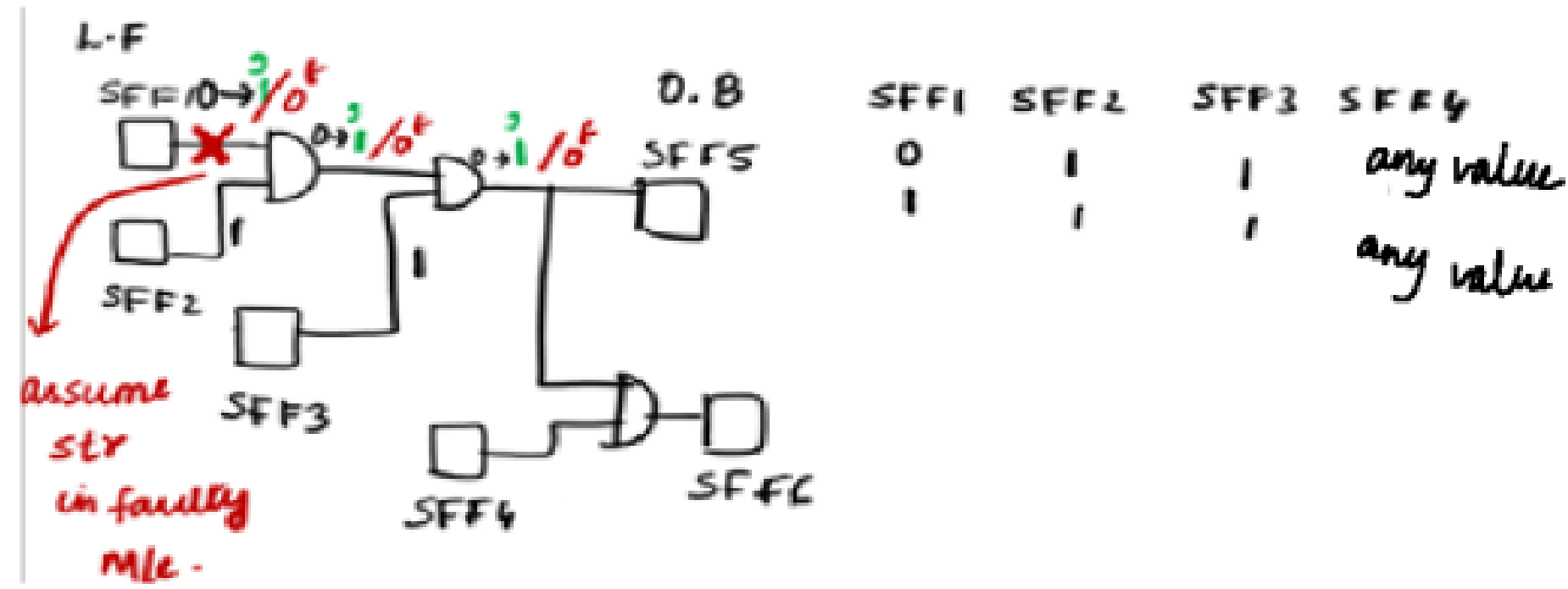
DIFFERENCE BETWEEN PATH DELAY FAULT MODEL AND TDF

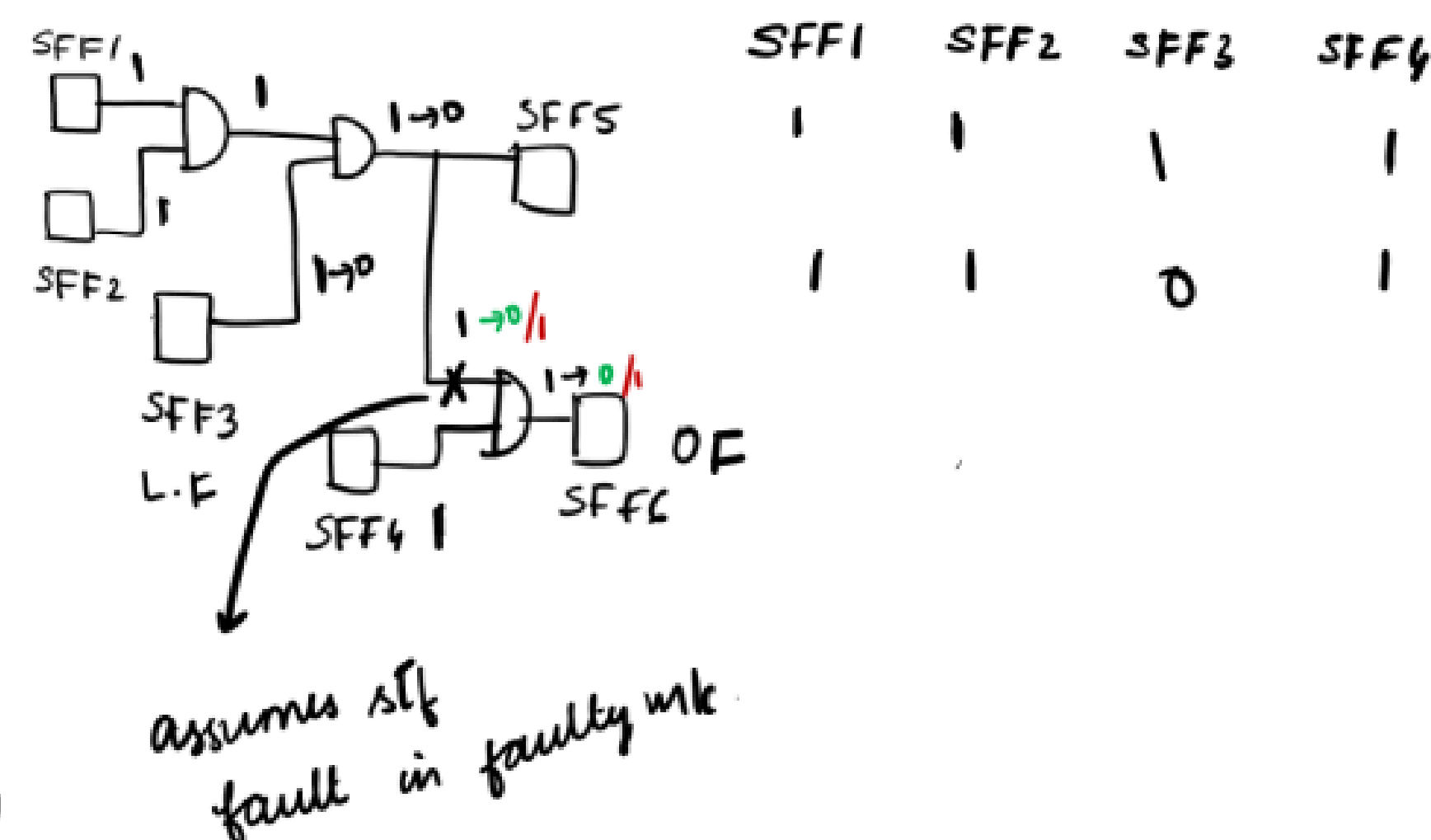
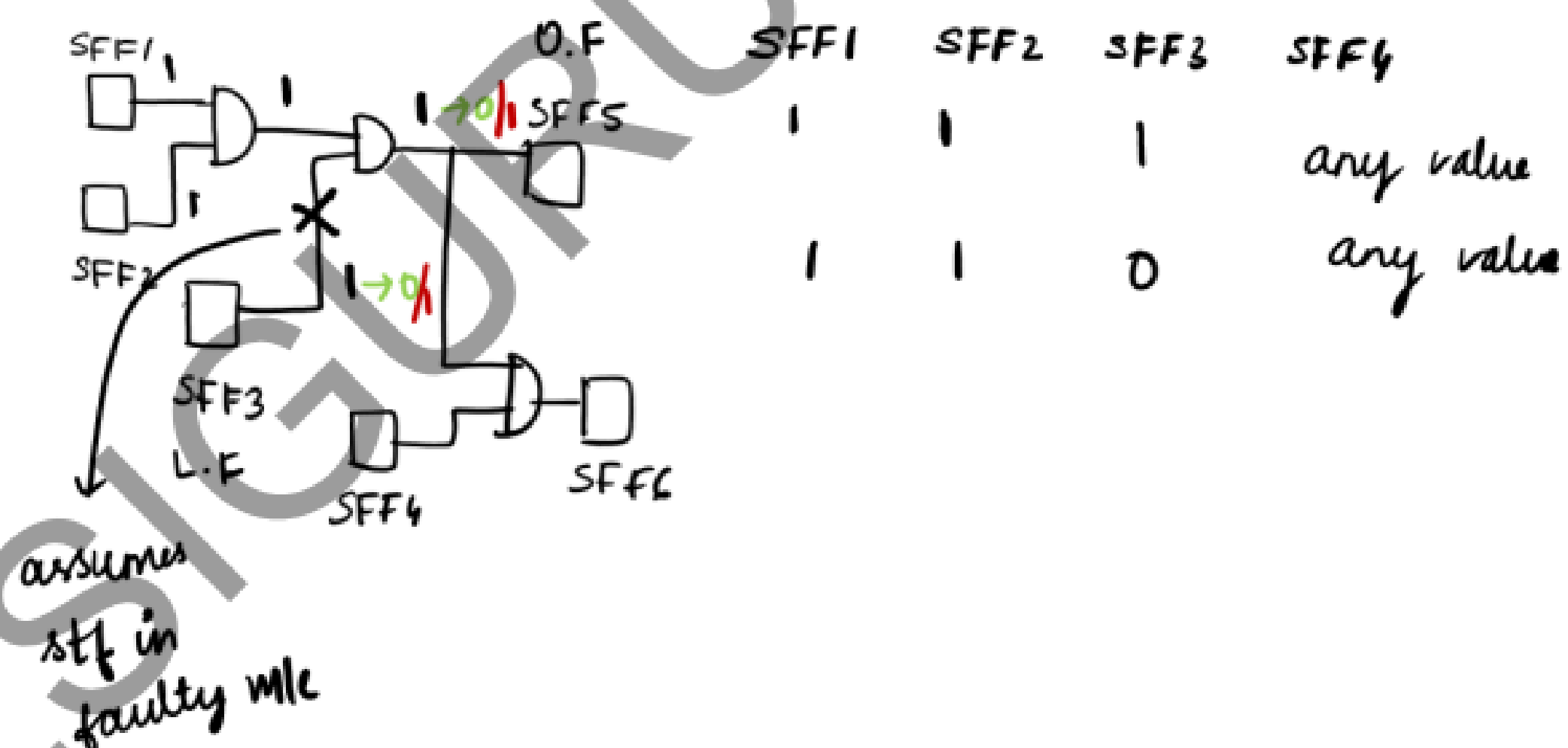
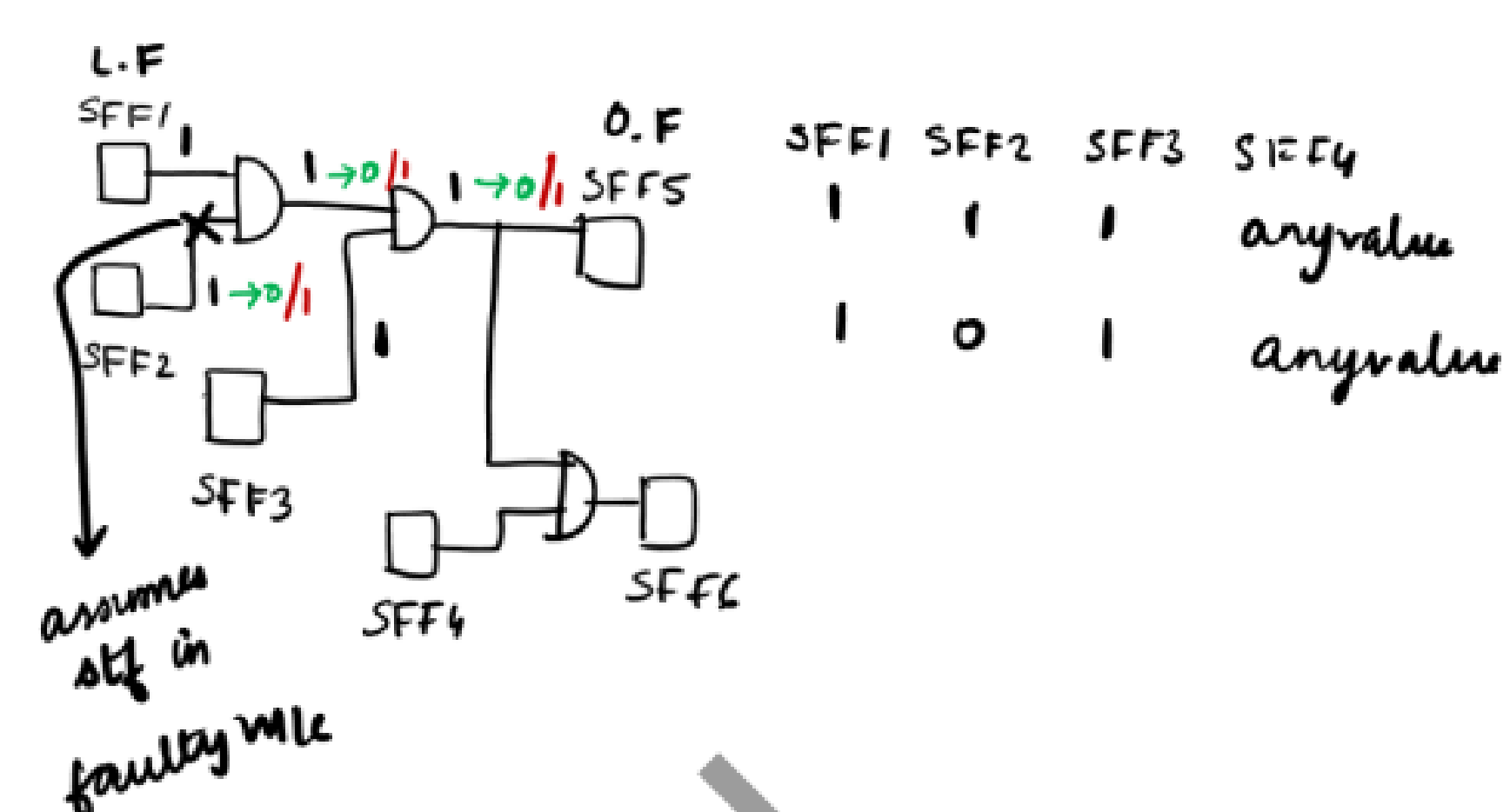
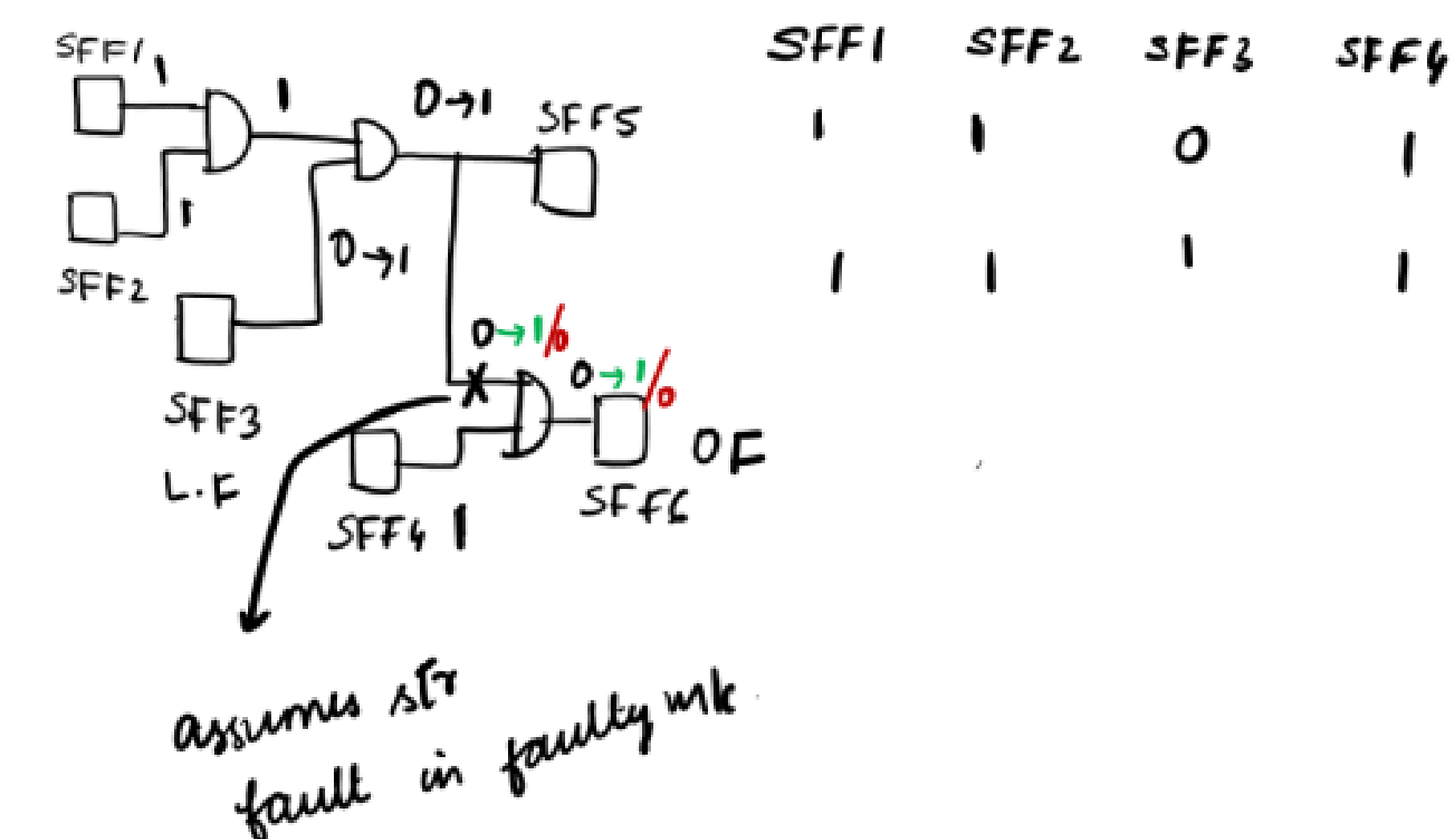
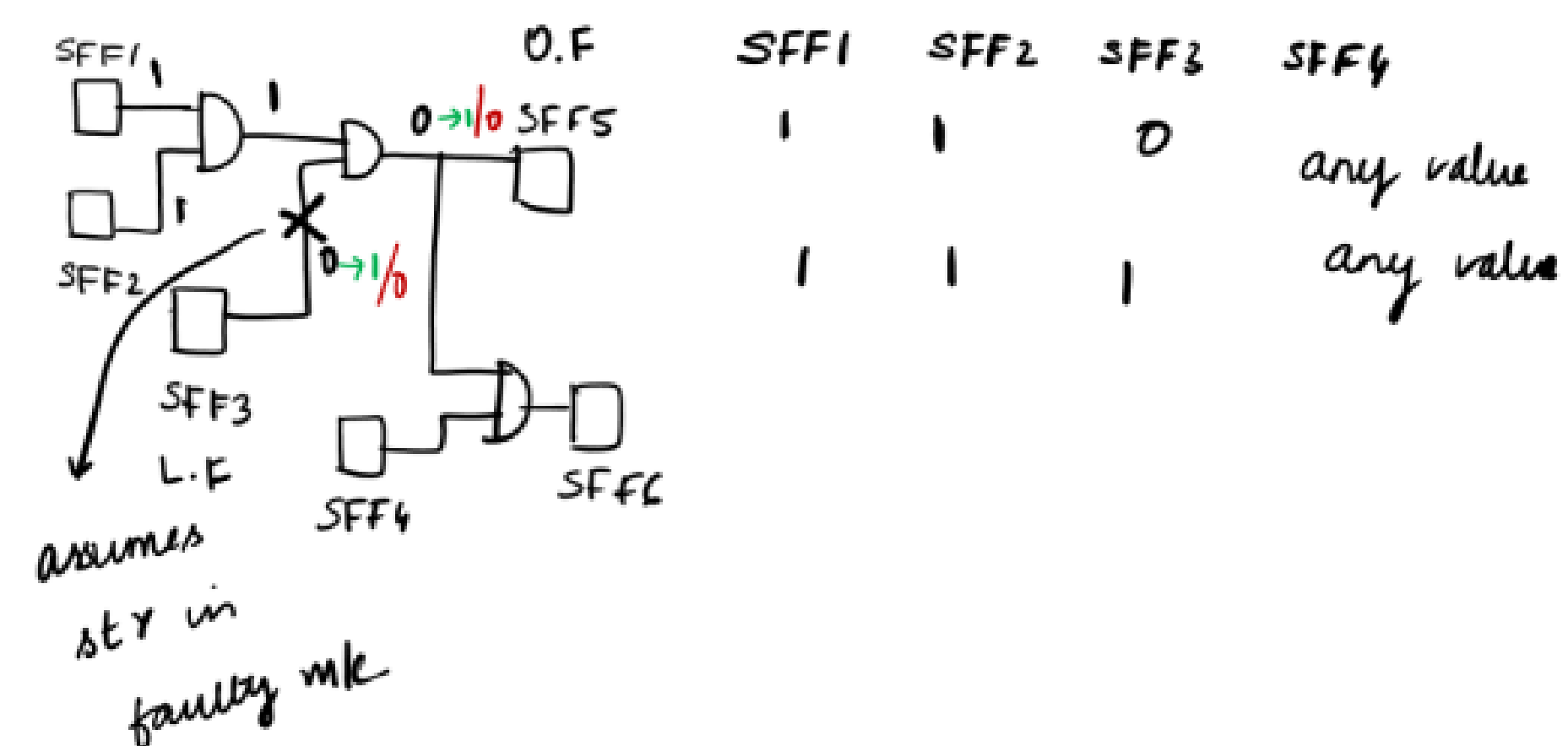
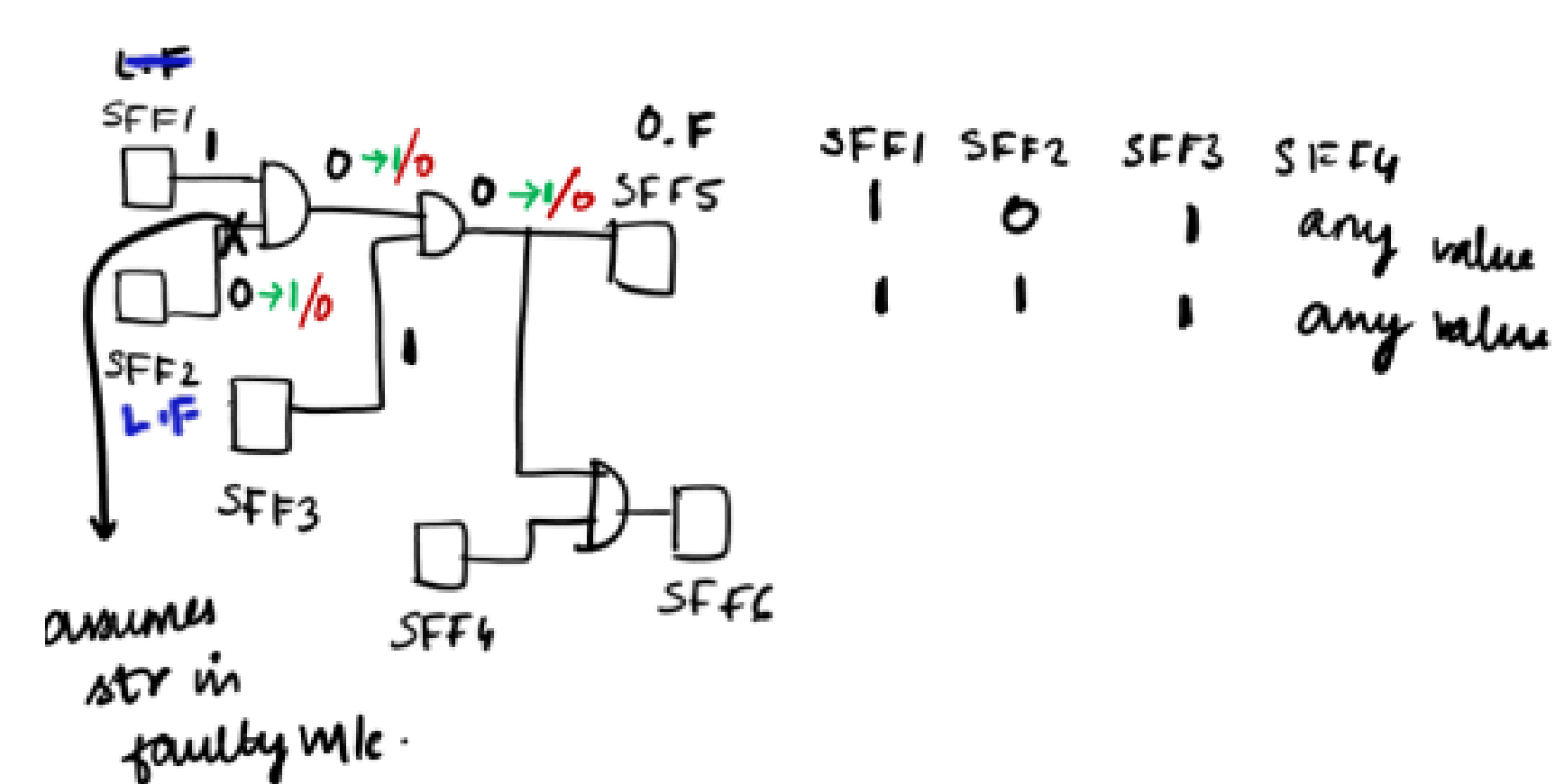
- **Path Delay Fault model** : Specific launch and observation point
- **Transition Delay Fault model** : no specific launch and observation point (selected by the tool). ATPG tool is free to determine the launch and capture points.

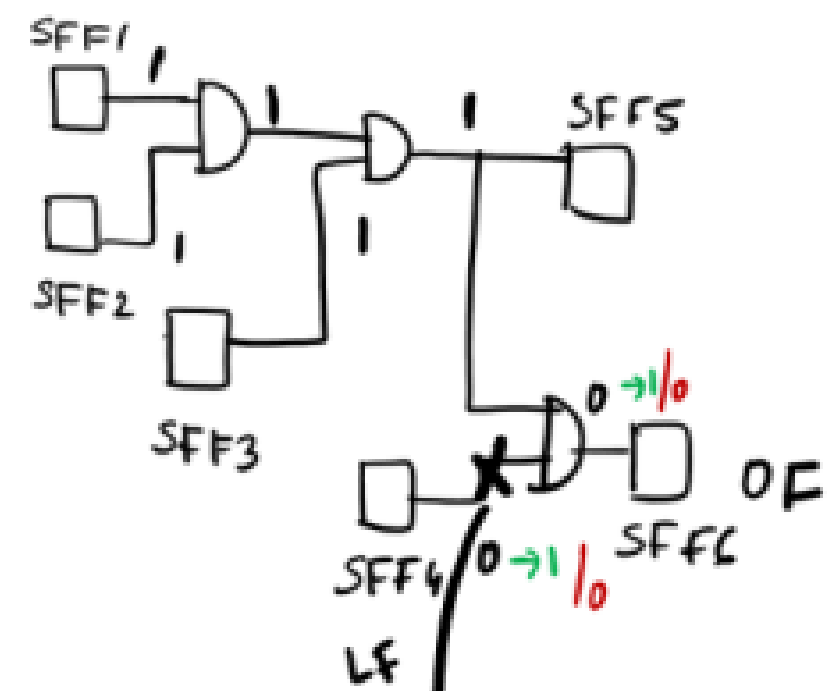
Example :-
Ckt :-



During ATPG TDF run process: Generates patterns using fault simulation

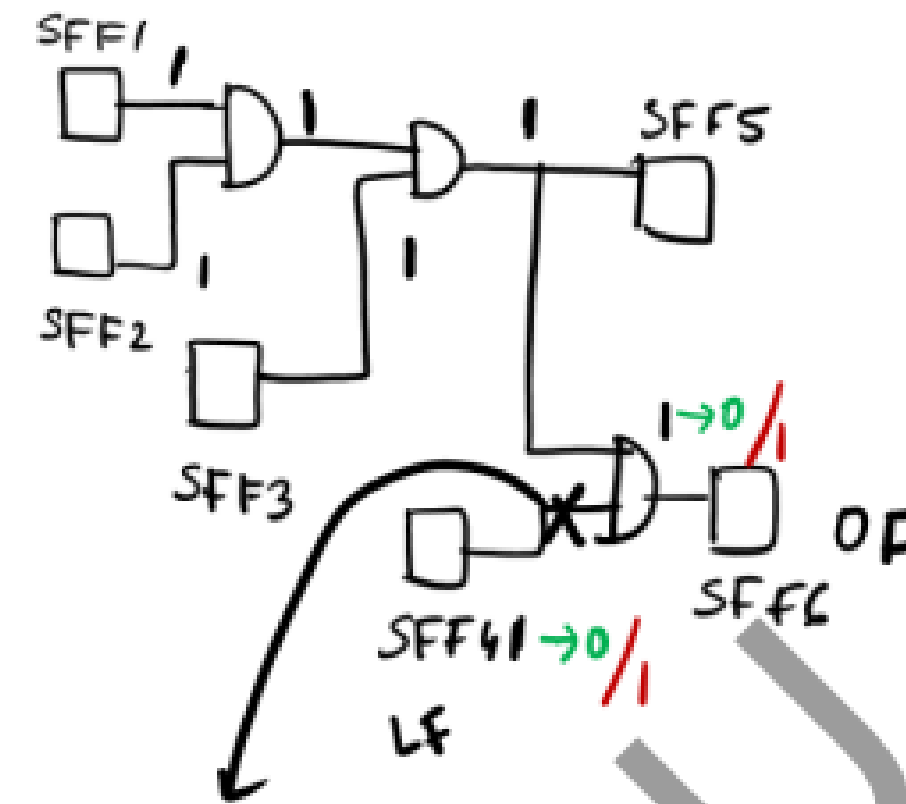






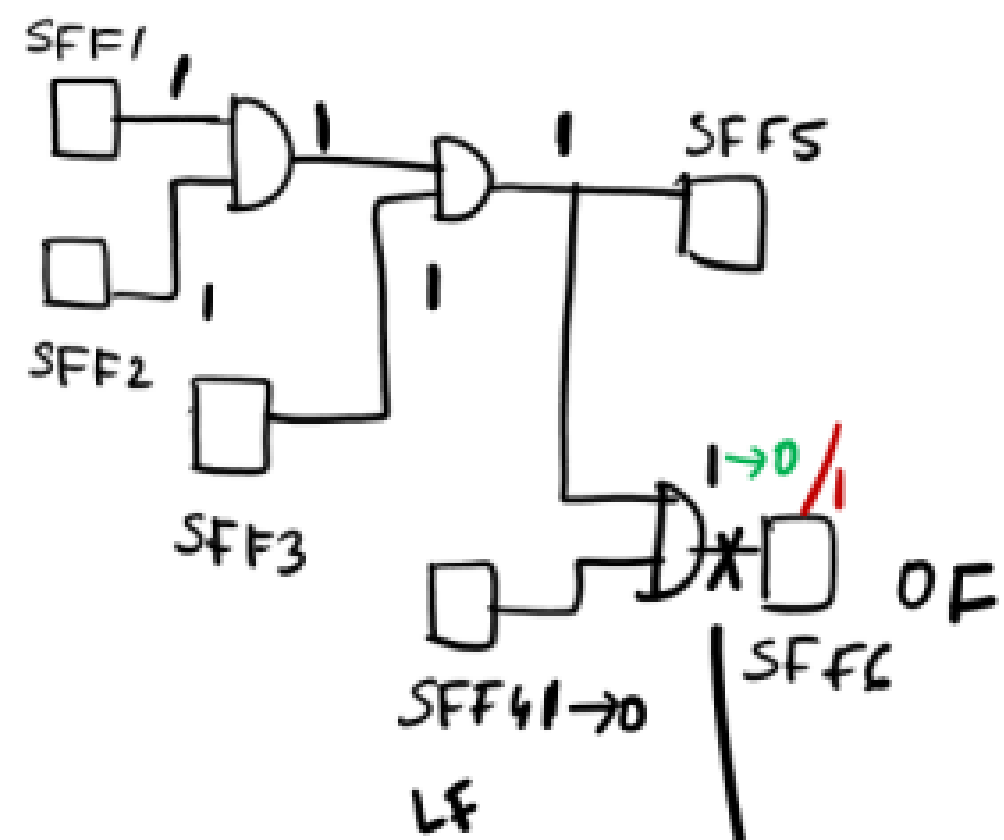
assumes
str in faulty
m/c

SFF1	SFF2	SFF3	SFF4
1	1	1	0
1	1	1	1



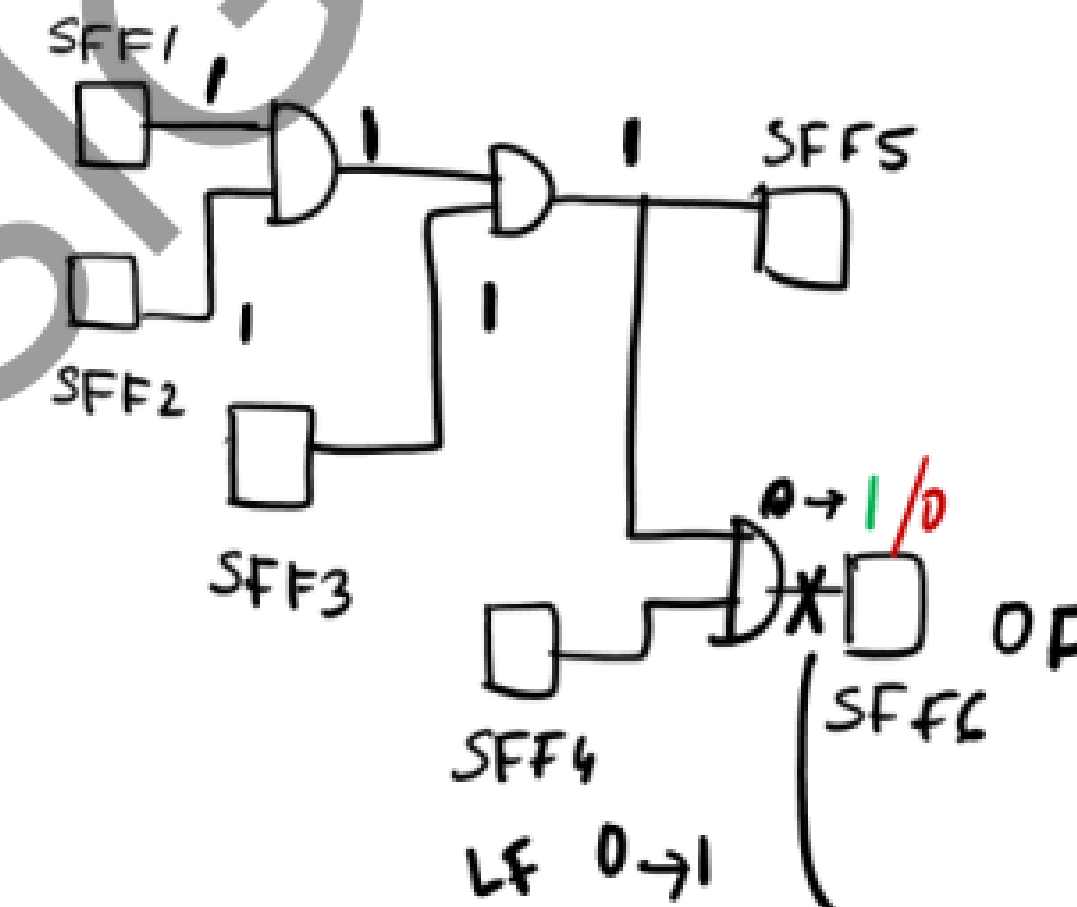
assumes
str in faulty m/c.

SFF1	SFF2	SFF3	SFF4
1	1	1	1
1	1	1	0



assumes str in faulty m/c.

SFF1	SFF2	SFF3	SFF4
1	1	1	1
1	1	1	0



assumes str fault
in faulty m/c.

SFF1	SFF2	SFF3	SFF4
1	1	1	0
1	1	1	1

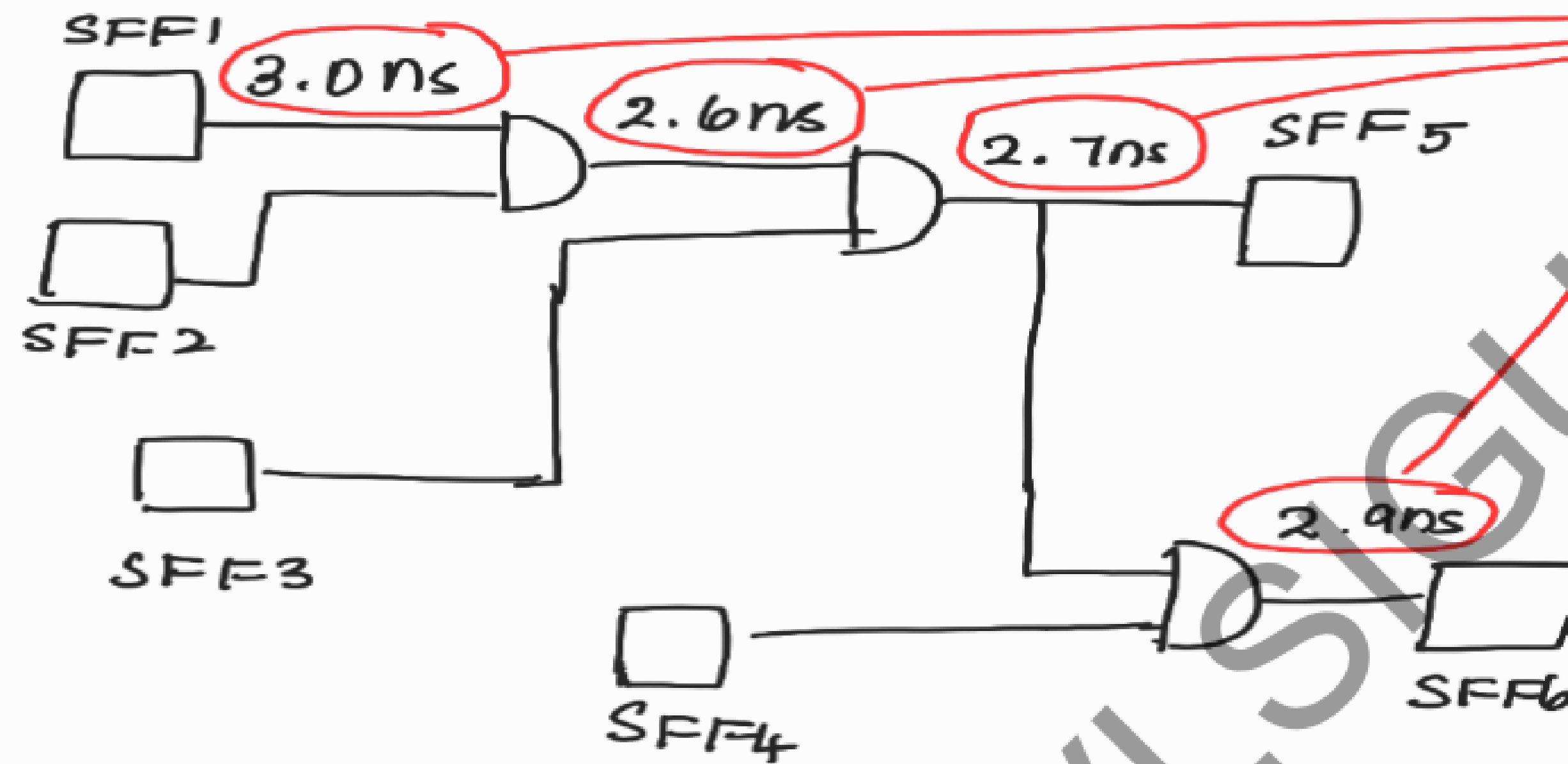
TO
O/P of ATPG $\Rightarrow$ Patterns $\Rightarrow$ Test the chip after manufacturing

Consider the following example :

Assume

clock period = 10ns

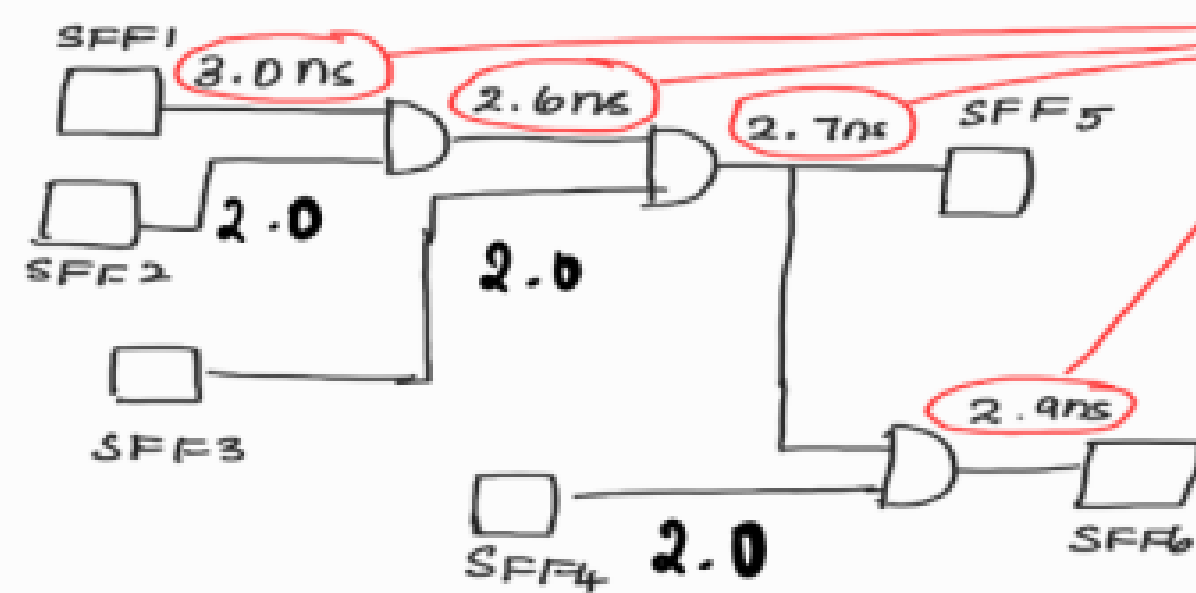
good cell + good net delay = 2.0ns



During manufacturing, due to some defects the delays are more compared to 2.0ns

Here the manufacturing defect is distributed throughout the path.

AFTER MANUFACTURING : Patterns gen during ATPG is used to test the chip



During manufacturing, due to some defects the delays are more compared to 2.0ns

clock period = 10ns
good cell + net delay = 2.0ns

P②

i/p seq

SFF1	SFF2	SFF3	SFF6
1	1	1	any
0	1	1	any

exp o/p

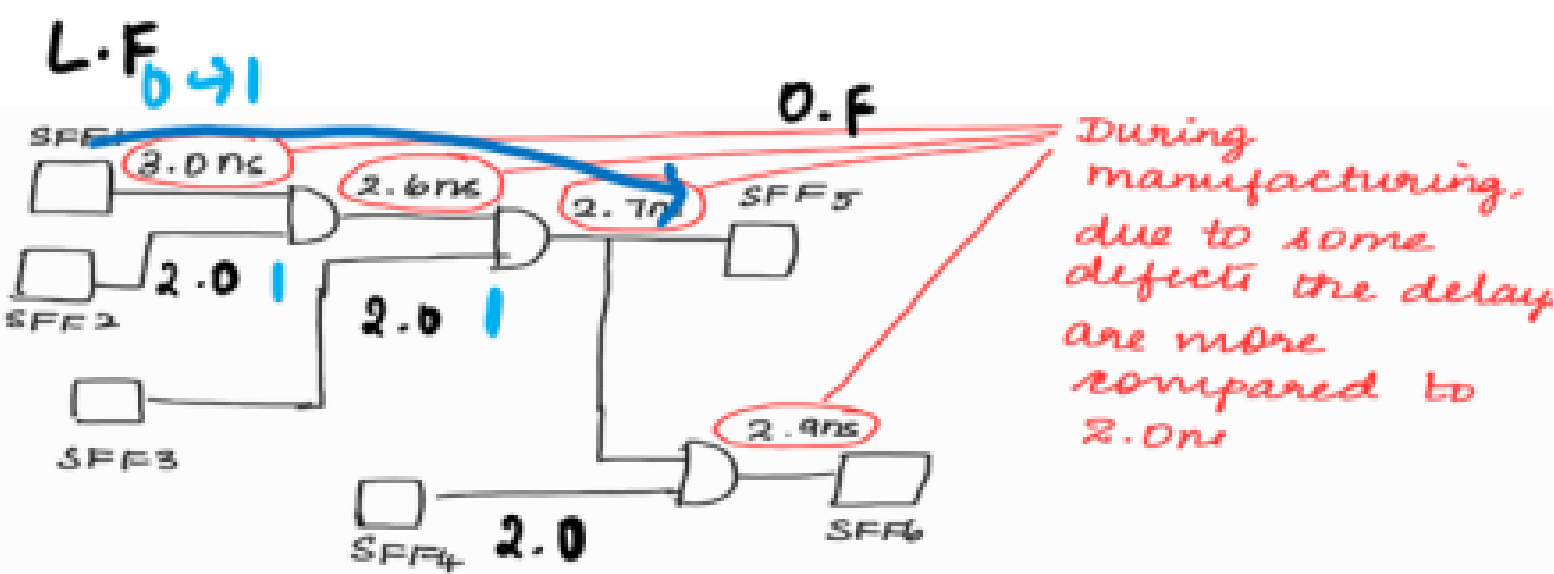
SFF5 = 0 after c pulse.

P①

i/p seq

SFF1	SFF2	SFF3	SFF4
0	1	1	any value
1	1	1	any value

exp o/p
SFF5 = 1 after c pulse.



During manufacturing, due to some defects the delays are more compared to 2.0ns

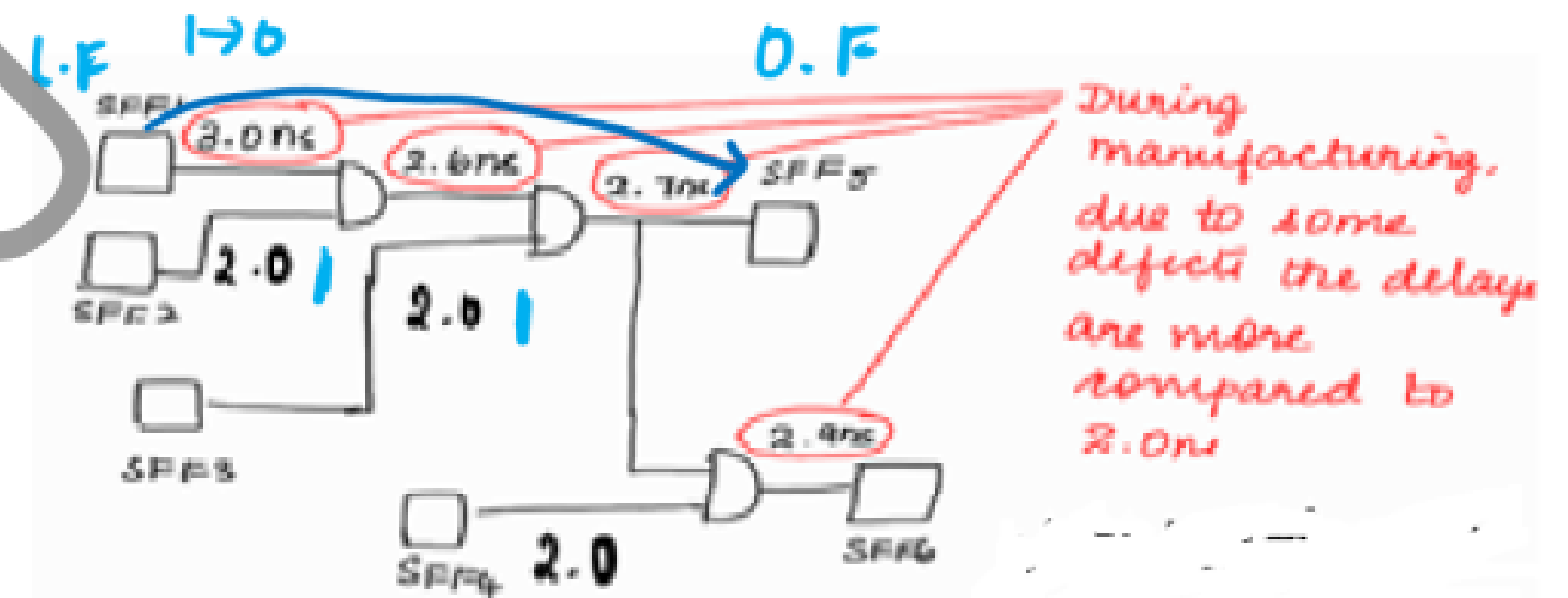
Time taken for transition
value to reach 0.F from L.F

$$3.0 + 2.6 + 2.7 = 8.3 < 10$$

∴ SFF5 will capture 1 as the transition has reached 0.F by the time a pulse comes (i.e., within 10ns)

∴ P1 is passed.

(∴ o/p of chip == exp o/p)



During manufacturing, due to some defects the delays are more compared to 2.0ns

Time taken for L value to reach 0.F from L.F

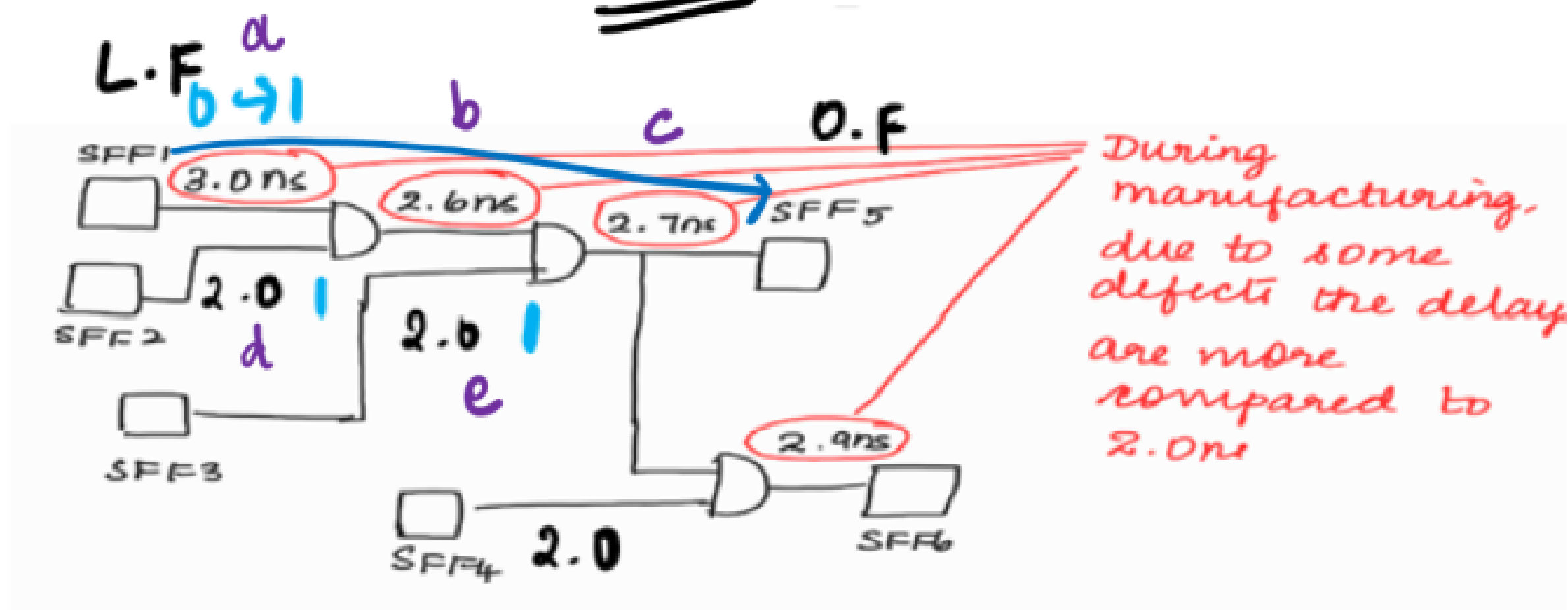
$$3.0 + 2.6 + 2.7 = 8.3 < 10$$

∴ SFF5 will capture 0

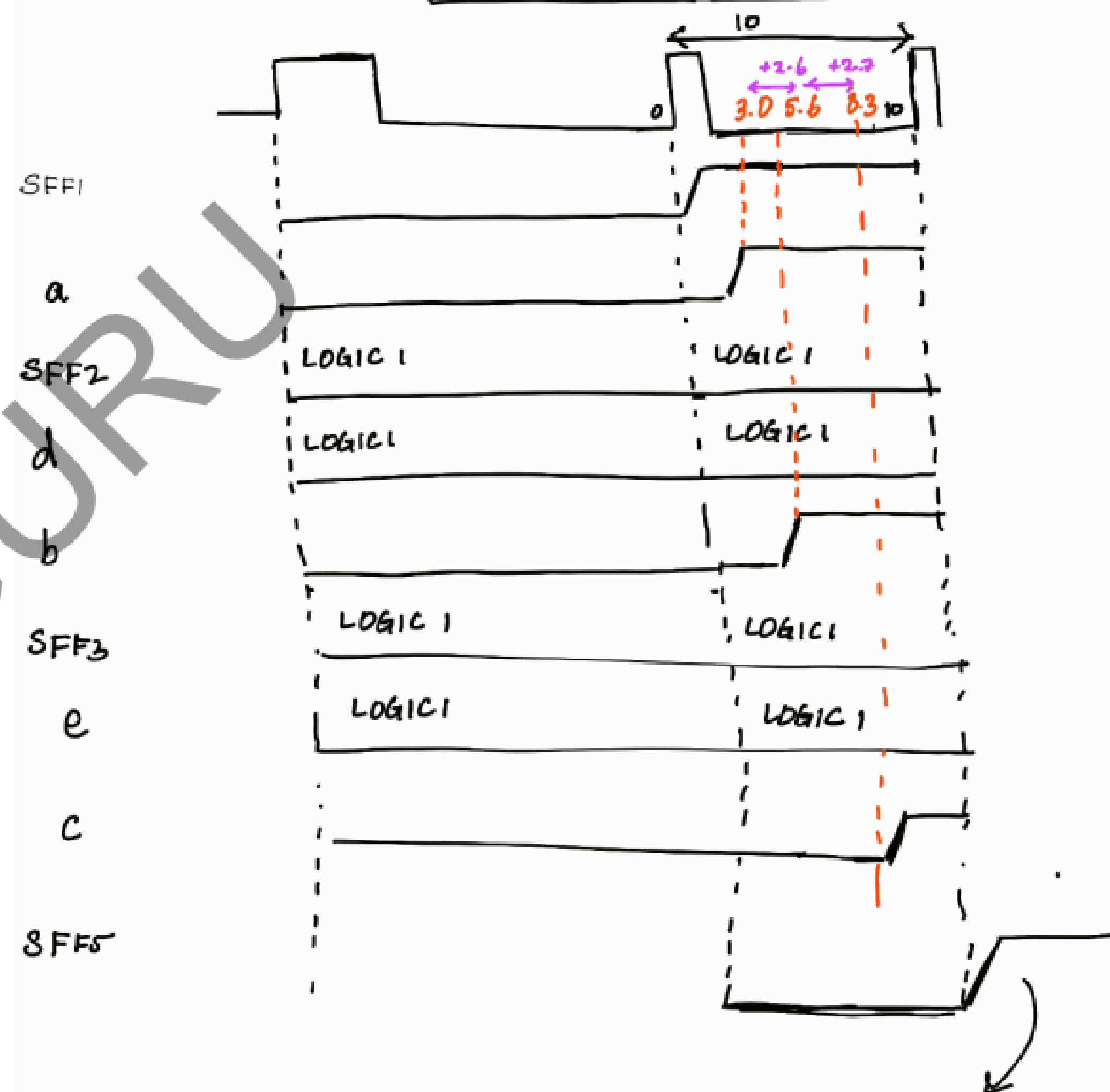
∴ P2 is passed.

(∴ o/p of chip == exp o/p)

NOTE:



SE



= = exp value

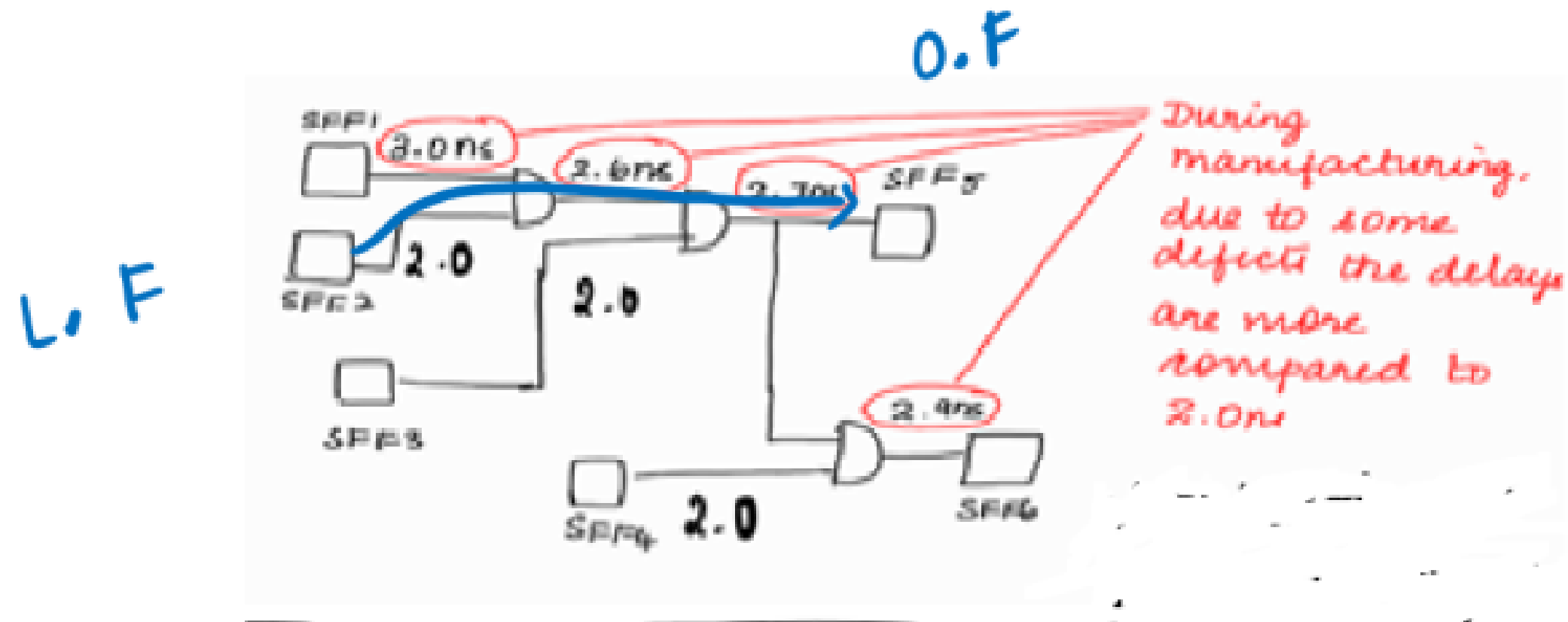
∴ Passed

P③

SFF1	SFF2	SFF3	SFF4
1	0	1	any value
1	1	1	any value

P④

SFF1	SFF2	SFF3	SFF4
1	1	1	any value
1	0	1	any value



Time taken for transition

value to reach 0.F from L.F

$$2.0 + 2.6 + 2.7 = 7.3 < 10$$

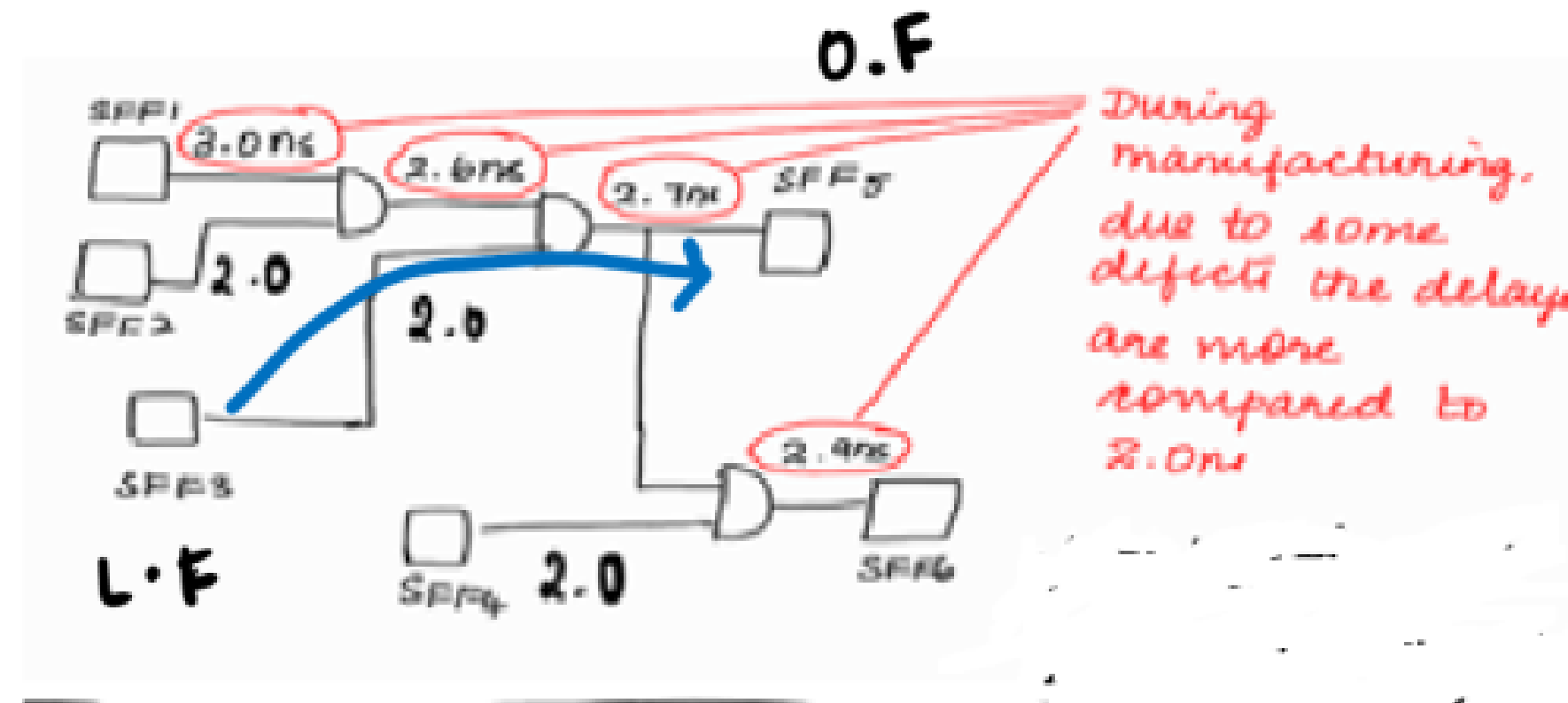
∴ P③ and P④ passed.

P5

SFF1	SFF2	SFF3	SFF4
1	1	0	any value
1	1	1	any value

P6

SFF1	SFF2	SFF3	SFF4
1	1	1	any value
1	1	0	any value



Time taken for transition
value to reach 0.F from L.F

$$2.0 + 2.7 = 4.7 < 10$$

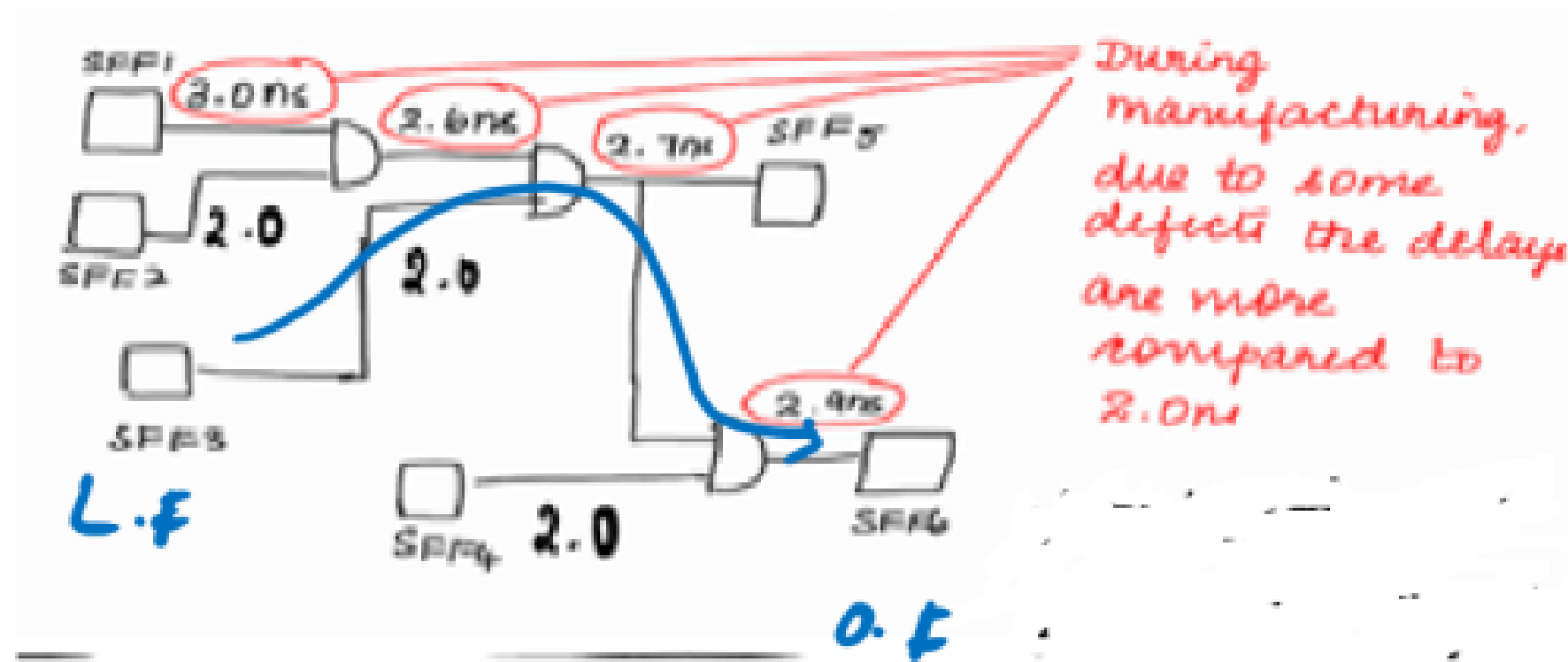
$\therefore P5, P6$ will pass

P7

SFF1	SFF2	SFF3	SFF4
1	1	0	1
1	1	1	1

P8

SFF1	SFF2	SFF3	SFF4
1	1	1	1
1	1	0	1



Time taken for transition value to reach

L.F to 0.F

$$2.0 + 2.7 + 2.9$$

$$= 7.6 < 10$$

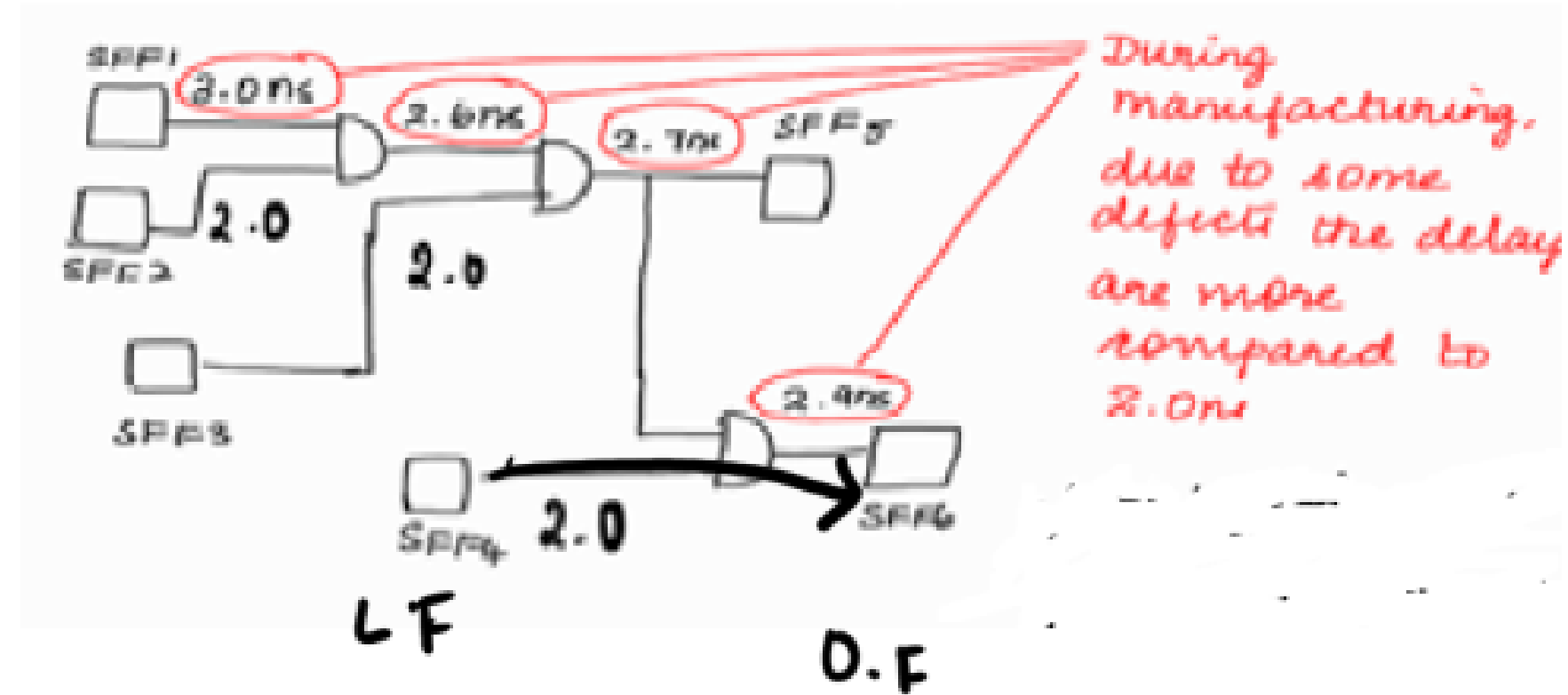
$\therefore$ P7, P8 will pass.

P(9)

SFF1	SFF2	SFF3	SFF4
1	1	1	0
1	1	1	1

P(10)

SFF1	SFF2	SFF3	SFF4
1	1	1	1
1	1	1	0



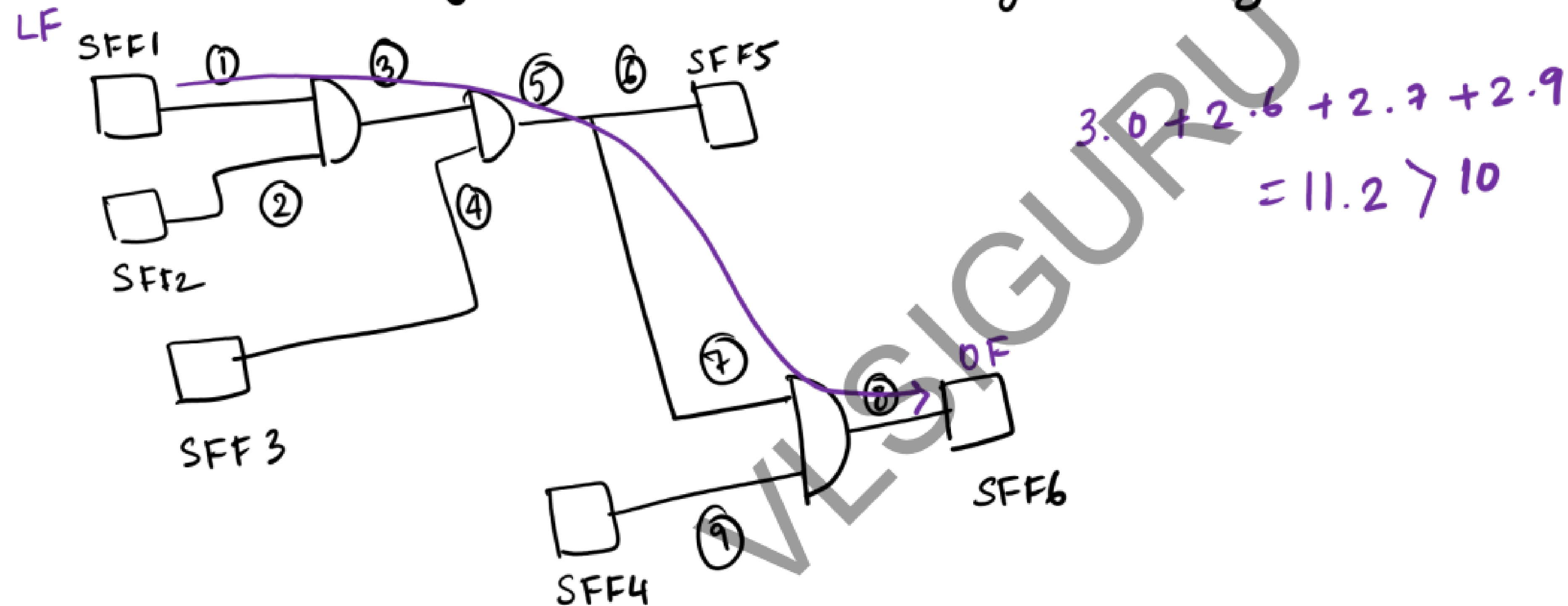
Time taken for the transition value to reach 0.F from LF

$$2.0 + 2.9 = 4.9 < 10$$

$\therefore$ P(9), P(10) will pass

All the TDF patterns are passed (as illustrated above)

But still if the chip is delivered to the customer, it will cause problem
Since the signal degrades a lot when it goes through this path (longer path)

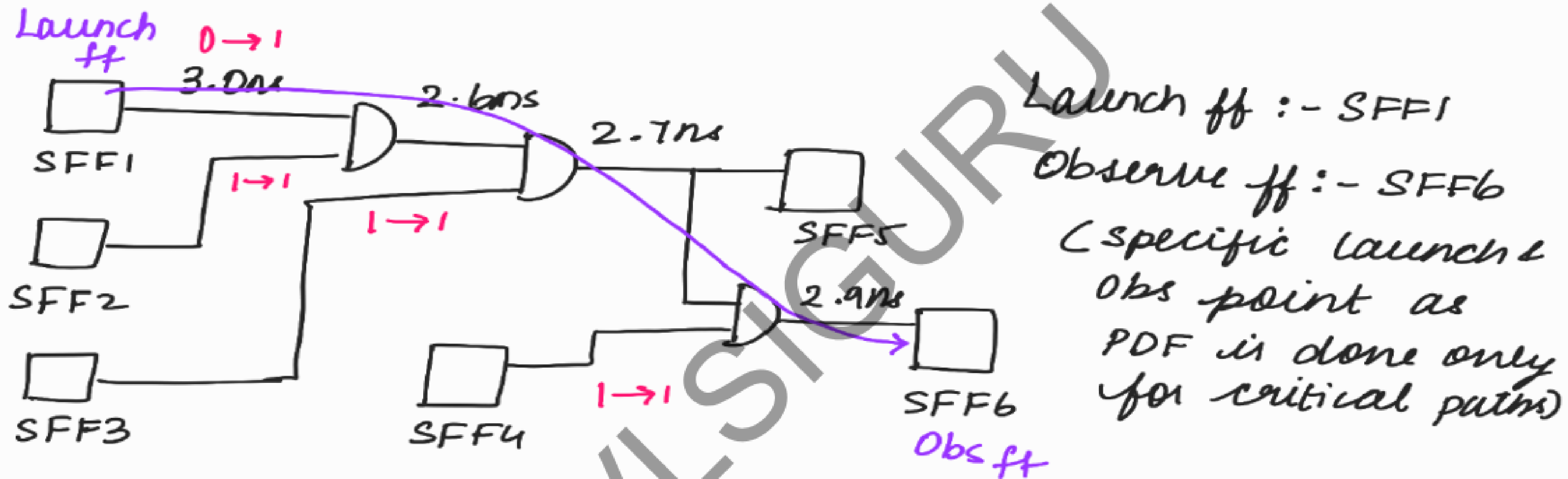


But this path was not tested in any of the TDF test patterns.

This is also a problem.

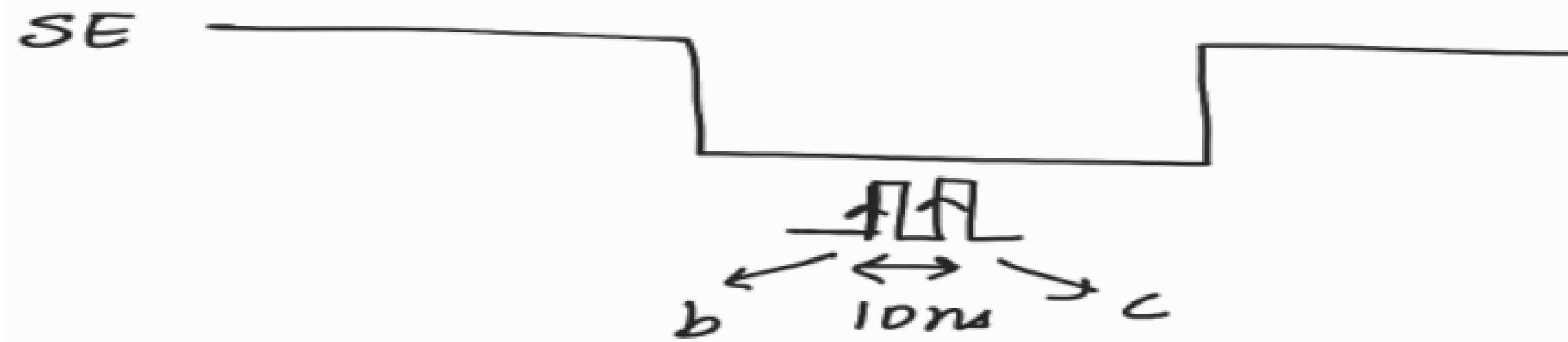
So we use PDF to test the critical paths

Path delay Fault model is associated with the testing of at-speed performance of critical path.



In this example, the path which is shown is the critical path ($\because$ it is the longest).

$$3.0ns + 2.6ns + 2.7ns + 2.9ns = 11.2ns > 10ns$$



Expected value
written by ATPG tool :-
 after c pulse, obs flop
 (SFF6)
 has to hold 1 value.

But after manufacturing, due to defects (as shown), transition value has not reached SFF6 by the time

c pulse has come (i.e. within 10ns transition has not reached SFF6. It has taken 11.2ns for the transition value to reach SFF6 from SFF1.)

So SFF6 will capture 0
 $\therefore$ test will fail.

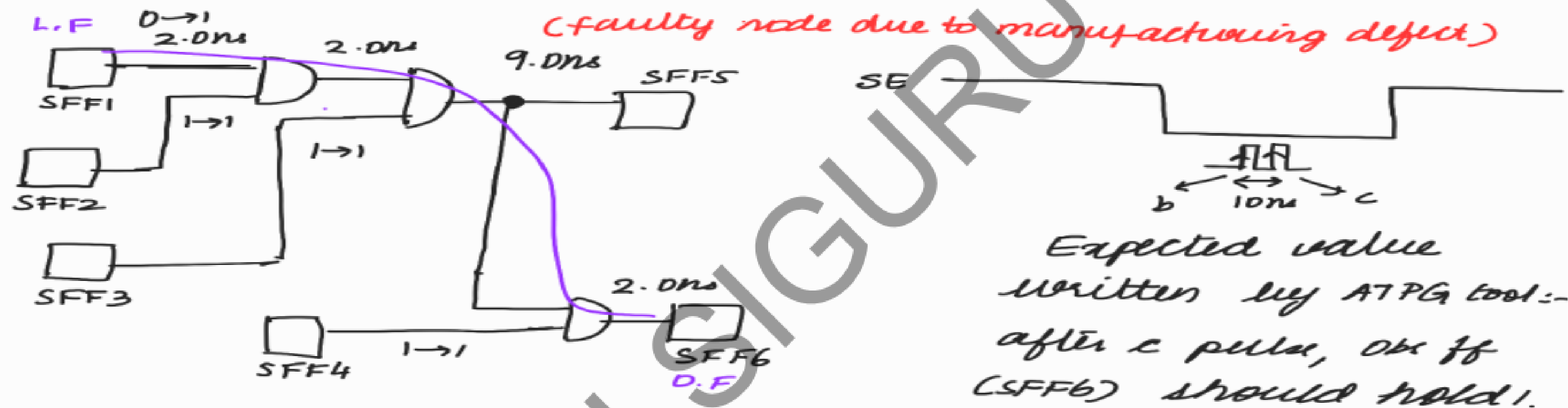
$\therefore$ the manufacturing defect is detected

↓
 manufacturing defect distributed
 throughout the path.

Transition Delay Fault Model can be used to detect manufacturing defect at a particular node.

Because the test will fail irrespective of the launch & obs points.

Eg:- Good cell + net delay = 2.0ns, clock period = 10ns



$$2.0ns + 2.0ns + 9.0ns + 2.0ns = 15ns > 10ns$$

By the time c pulse comes (i.e. within 10ns), transition value didnot reach SFF6.

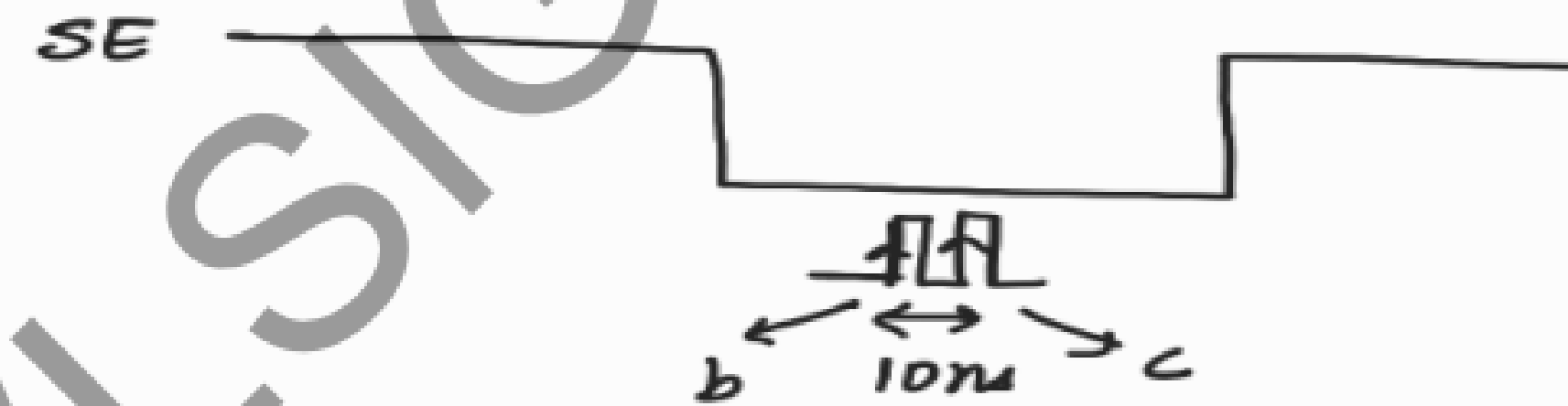
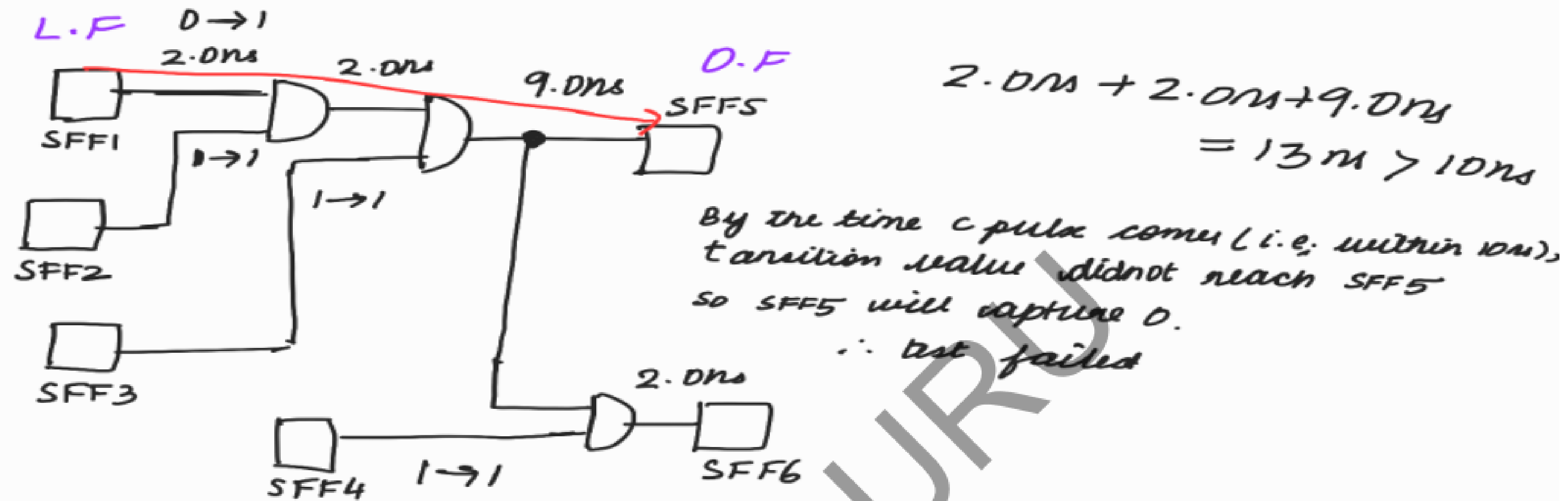
So SFF6 will capture 0.

∴ test failed

NOTE:- FOR TDF

there is no specified launch or observation point.

ATPG tool is free to determine the launch, capture points.



Expected value
 written by ATPG tool:
 after c pulse, 0x ff
 (SFF5) should hold 1.

We need both PDF and TDF because, with only TDF, we can't assure that manufacturing defect which is distributed throughout the path will get detected. So we usually do PDF to test critical path.

Run-time of PDF is high. So we usually do PDF to test at-speed performance of critical paths.

To test the other paths, we use TDF.